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يُسر- Yusr

Startup Business Plan

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1 Executive Summary

Every year, millions of Hajj and Umrah pilgrims face significant challenges in managing rituals, staying with their groups, and maintaining their health in extremely crowded and physically demanding environments. In 2024 alone, over 1,300 pilgrims died due to heat-related causes [1], highlighting the urgent need for safer, real-time support solutions.

Yusr is a smart wearable system designed specifically for pilgrimage experience. It combines an intelligent wristband with a companion mobile application to provide hands-free, real-time support. The system automatically tracks Tawaf and Sa'i rituals using AI-powered motion sensors, monitors vital health indicators, and enables instant emergency alerts with location sharing. Our landing page and explainer video MVPs have already driven early user registrations and confirmed strong demand for a hands-free, integrated pilgrimage support solution.

The key value of **Yusr** lies in its fully integrated, hands-free approach, allowing pilgrims, especially elderly and high-risk individuals, to focus on their worship without relying on mobile phones. Unlike existing mobile-based solutions such as Pilgrim App, Tawaf Counter, and Nusuk, which remain impractical during active rituals and lack integrated wearable safety capabilities, **Yusr** is the only solution that simultaneously addresses ritual accuracy, health monitoring, and group safety in a hands-free format.

Yusr targets Hajj and Umrah pilgrims of all ages and nationalities, offered through direct purchase at 267 SAR and short-term rental starting at 40 SAR per day, within a Serviceable Obtainable Market of approximately 599 million SAR, with a first-year target of 47,885 units sold and 428,926 units rented. We project first-year revenue of 29.9 million SAR, with a financial plan targeting break-even by year four.

The solution has been validated through 53 surveys and 50 interviews with pilgrims. Findings revealed that 77.4% of pilgrims struggled to track companions in crowds, 71% feared losing a family member, and over 93% expressed willingness to use a smart bracelet during pilgrimage, with a clear preference for wearable, hands-free solutions over traditional mobile applications.

2 Product/Service Description

2.1 Problem & Solution

Yusr addresses key challenges faced by Hajj and Umrah pilgrims: inaccurate ritual counting, health risks caused by physical strain and extreme heat, and difficulties in maintaining group connectivity in crowded environments. Through an innovative system consisting of a smart wristband integrated with a dedicated mobile application, **Yusr** delivers a hands-free, real-time support platform designed to significantly reduce these challenges entirely.

2.2 Main Features

Yusr consists of a **smart wristband** paired with a **companion mobile application**, offering six core features:

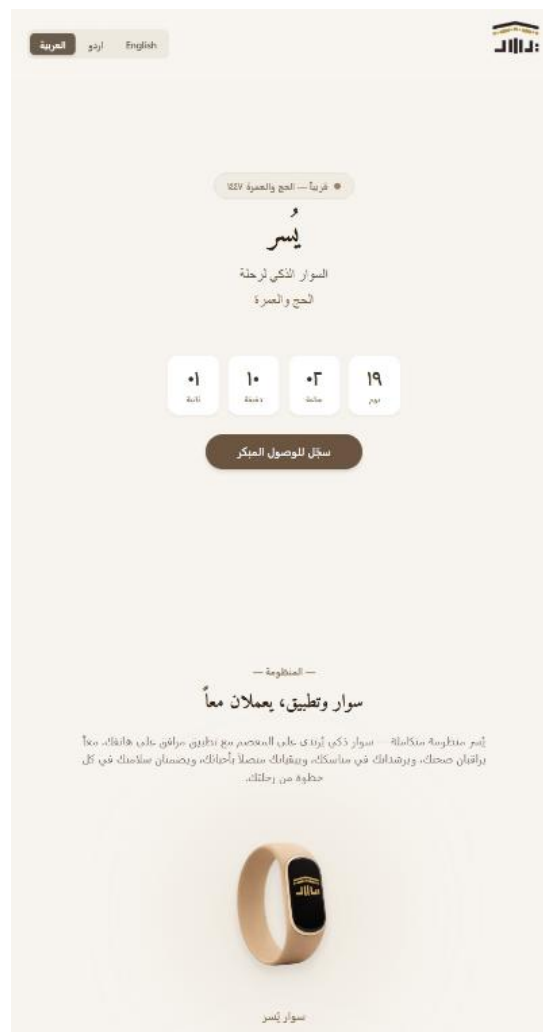
- **Automatic Ritual Counting:** AI-powered motion sensors detect and count Tawaf and Sa'i rounds without GPS or manual input, eliminating common sources of ritual error.
- **Health Monitoring:** Continuously tracks vital signs and triggers personalized alerts when thresholds are exceeded, reducing the risk of heat exhaustion and fatigue.
- **SOS Emergency System:** Automatically transmits the pilgrim's precise location and health data to authorized contacts and supervisors, enabling rapid response.
- **Voice Communication:** Direct voice calls from the wristband remove the need to access a phone during rituals.
- **Companion Mobile App:** Provides step-by-step ritual guidance, supplications, Grand Mosque gate map, and navigation support in Arabic, English, and Urdu.
- **Real-Time Location Tracking:** Displays the pilgrim's live location alongside detailed maps of the holy sites, enabling seamless navigation throughout the pilgrimage.

- **Communication Centre:** Facilitates easy message sending and receiving of broadcasts from authorized contacts, ensuring pilgrims stay informed and connected.
- **Gate Map Navigation:** Provides detailed gate maps of the Grand Mosque to help pilgrims navigate entrances and exits with confidence.

2.3 Minimum value product (MVP):

To validate **Yusr's** concept prior to hardware development, a lightweight MVP was created consisting of two key components: a multilingual **landing page** and an **explainer video**.

The landing page, available in Arabic, English, and Urdu, introduces the system and highlights its core features, previews the mobile application interface (as shown in Figure 1), and includes an early access registration form to capture user interest. A call-to-action button is strategically placed throughout the page to enable users to act quickly when convinced (as shown in Figure 1). You can visit our landing page through this link: <https://yusr.base44.app/>



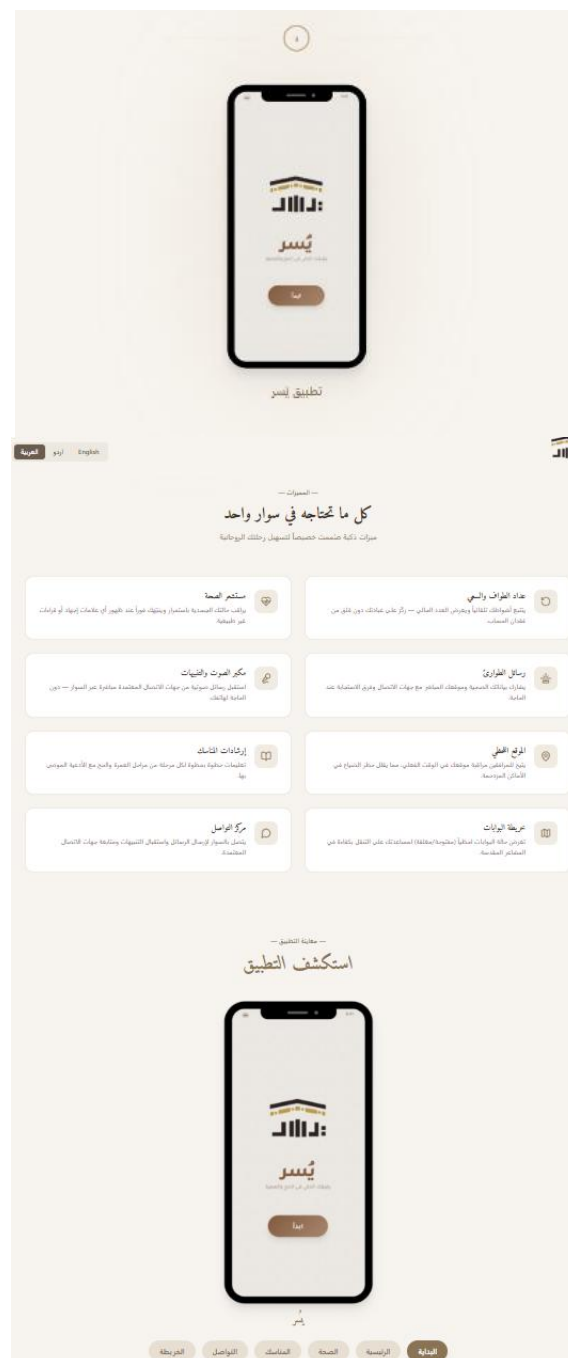


Figure 1: Yusr's Landing Page

Additionally, a simple registration form allows users to sign up and be among the first to experience **Yusr** when it launches (as shown in Figure 2).

Figure 2: Early Access Registration Form

Complementing the landing page, an explainer video visually demonstrates the real-life challenges faced by pilgrims, such as losing count during Tawaf, experiencing heat fatigue, and separation from companions, before presenting **Yusr** as a practical solution. The video features Arabic narration with English and Urdu subtitles and ends with a call-to-action directing viewers to the landing page (as shown in Figure 3). You can watch our explainer video through this link: <https://youtu.be/r3xUVeEphLI?feature=shared>

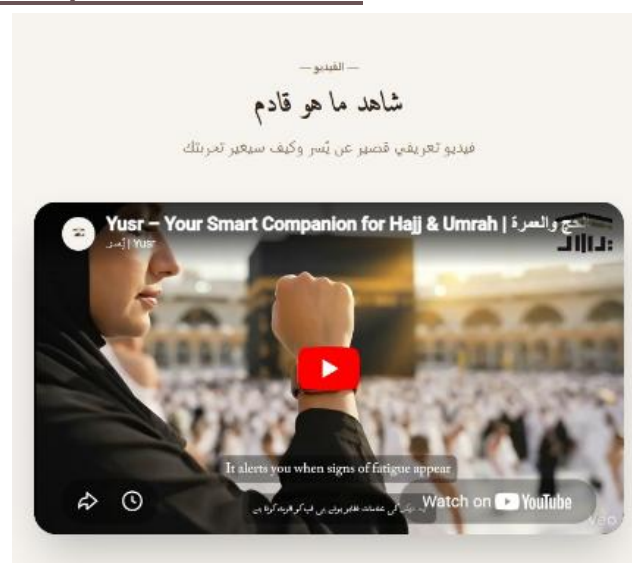


Figure 3: Yusr's Explainer video

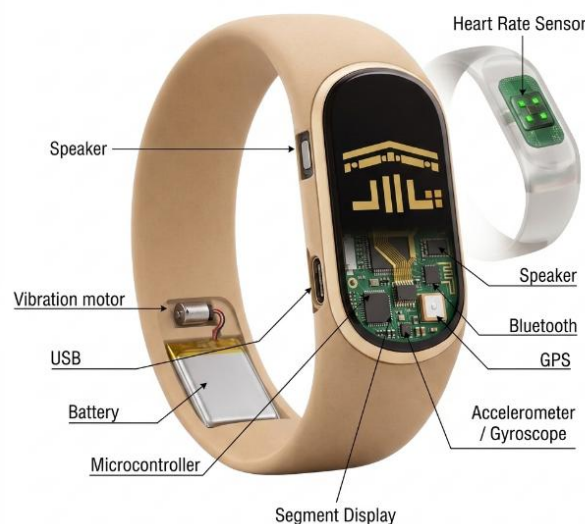
2.4 Envisioned Full Product

In its fully developed form, **Yusr** will evolve into a comprehensive smart pilgrimage ecosystem. The wristband will incorporate advanced sensors with improved accuracy and extended battery life, while the mobile application will deliver AI-driven health insights, predictive alerts, and enhanced navigation capabilities. The system is designed to support large-scale Hajj campaign deployments and potential integration with the Ministry of Hajj and Umrah, enabling real-time crowd monitoring, faster emergency response, and a significantly improved pilgrimage experience for every pilgrim worldwide.

The fully developed **Yusr** system will expand to include:

- **Advanced Ritual Counting:** Enhanced AI accuracy for detecting Tawaf and Sa'i rounds in highly crowded environments.
- **Predictive Health Alerts:** AI-driven analysis of vital signs to predict and prevent health incidents before they occur.
- **Crowd Monitoring Integration:** Real-time crowd density mapping shared with Ministry of Hajj and Umrah supervisors.
- **Extended Battery Life:** Improved power management for multi-day Hajj usage without recharging.
- **Enhanced Navigation:** Detailed indoor mapping of Masjid Al-Haram with real-time guidance.

The following figures illustrate the design of **Yusr's** wristband and mobile application.



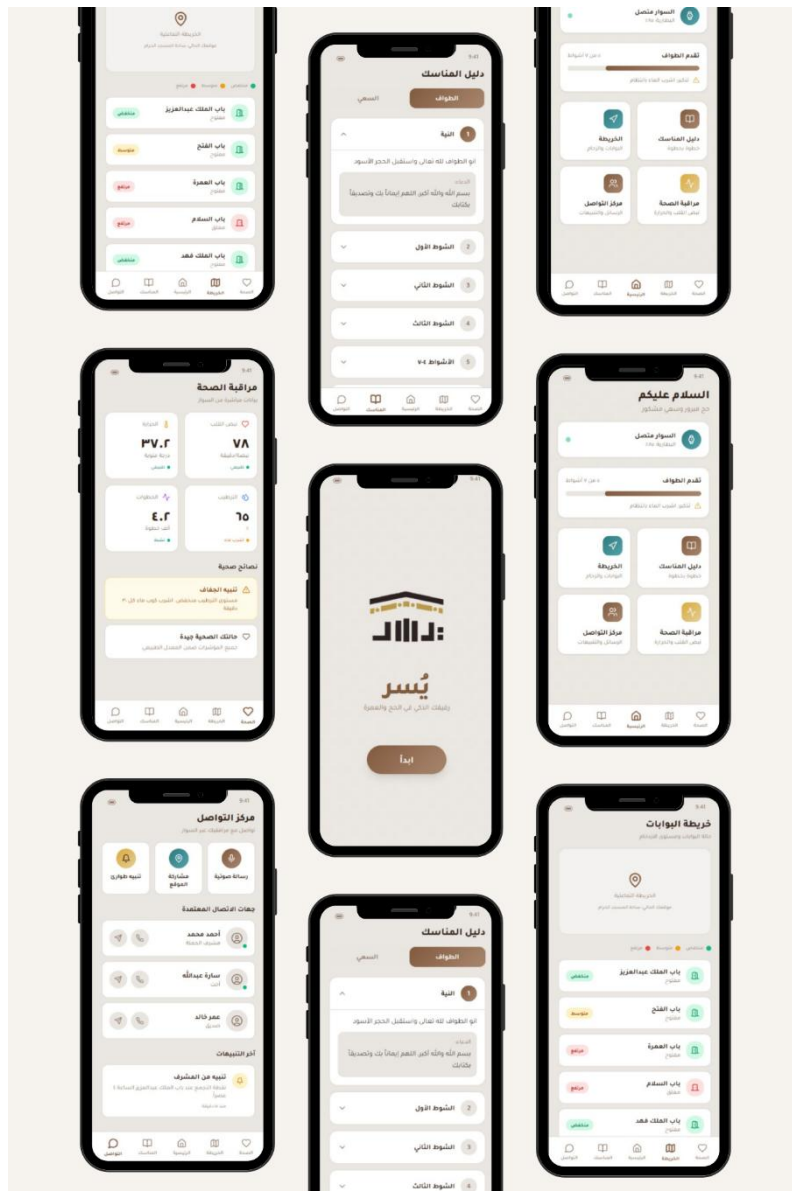


Figure 4: Yusr's Wristband and App

3 The Team

At **Yusr**, our strength lies in our team. We are six motivated individuals with diverse skills in technology, design, and business, working together to build a smart solution that improves the Hajj and Umrah experience. United by a shared mission, each member plays a key role in developing and growing **Yusr**.

Below are an overview of each team member and the essential role they play in fueling the growth of our startup:

Name	Position	Work Experience	Qualification	Responsibilities
Fajer Alamro	Operations & Finance Manager	Fajer brings over three years of experience as an Operations & Finance Manager, specializing in financial planning, operational efficiency, and cross-functional collaboration to drive business growth	Fajer holds a bachelor's degree in finance and has earned professional certifications including CFA Level I and the Financial Modeling & Valuation Analyst (FMVA) certification.	As Operations & Finance Manager, Fajer oversees Yusr's financial operations and internal processes, ensuring efficiency, financial accuracy, and alignment with the company's strategic goals
Rama Alomair	AI Specialist	Rama brings over two years of experience in software development and AI, with a focus on building intelligent, scalable systems and data-driven solutions centered around user needs.	Rama holds a bachelor's degree in information technology and has professional certifications in Artificial Intelligence, including AI Specialist, Machine Learning, and Data Analytics.	As AI Specialist, Rama leads the development of Yusr's AI-driven features, ensuring the platform leverages smart, reliable solutions that align with user needs and business goals.
Aljawharah Alsubaie	UX/Industrial Designer & App UI Lead	Aljawharah has extensive experience in UX and industrial design, specializing in intuitive interfaces, user journeys, and seamless experiences across digital and physical products	Aljawharah holds a bachelor's degree in industrial design and professional certifications including Google UX Design Certificate, and Adobe Certified Professional in UI/UX Design	As UX/Industrial Designer & App UI Lead, Aljawharah leads the design of Yusr's user interfaces and user journey, ensuring the platform is intuitive, engaging, and easy to use

Rima Alsonbul	Hardware & Embedded Engineer	Rima has hands-on experience in hardware and embedded systems, specializing in sensor integration, firmware development, and hardware-software interface design	Rima holds a bachelor's degree in electrical engineering and professional certifications including Embedded Systems Essentials, ARM Cortex-M Embedded Systems Certification, and IoT Wireless & Cloud Computing Certification	As Hardware & Embedded Engineer, Rima leads the development of Yusr's hardware and embedded systems, ensuring seamless integration between the wearable device, sensors, and app
Remas Alsubaie	Marketing & Sales Manager	Remas has 2 years of experience in marketing and sales, focusing on brand growth, customer engagement, and campaign strategy	Remas holds a bachelor's degree in marketing and professional certifications including Google Digital Marketing & E-commerce Certificate, and Meta Certified Digital Marketing Associate	As Marketing & Sales Manager, Remas leads Yusr's marketing and sales efforts to grow brand awareness, engage customers, and drive business growth
Walaa Saifalislam	Chief Executive Officer (CEO)	Walaa has 4+ years of experience in leadership and project management. As CEO, she leads Yusr's vision and ensures effective execution of its business goals.	The CEO holds a bachelor's degree in business administration and professional certifications including Project Management Professional (PMP), Strategic Leadership and Management Certification, and Entrepreneurship in Emerging Economies (Harvard)	As CEO, Walaa leads Yusr's overall direction, ensuring clear strategy and effective execution of the company's goals

Table 1: Yusr Team

4 Market and Competitive Analysis

4.1 Target Customers

The target customers for **Yusr** are individual consumers within the B2C (Business-to-Consumer) segment. Demographically, **Yusr** targets Hajj and Umrah pilgrims of all ages and genders, spanning diverse nationalities and educational backgrounds. They are characterized by a readiness to embrace assistive technology for safety and spiritual purposes. In 2024, over 18.5 million pilgrims visited the Kingdom — 16.9 million for Umrah and 1.61 million for Hajj — with Saudi Arabia targeting 30 million Umrah pilgrims annually by 2030 [2]. Psychographically, **Yusr's** customers are spiritually motivated individuals who prioritize safety, peace of mind, and uninterrupted worship. They seek solutions that reduce logistical burdens and allow them to focus fully on their religious experience. In terms of buying patterns, our target consumers research pilgrimage-related products through online platforms, community recommendations, and travel agencies, and are willing to invest in wearable technology that demonstrates clear safety and convenience benefits.

We have chosen these customers because their shared goal of a safe, meaningful pilgrimage experience makes them highly aligned with the core value proposition of **Yusr**. A large proportion of pilgrims have chronic conditions, and the physical demands of Hajj can put them at significant health risk, underscoring the urgent need for real-time health monitoring during the pilgrimage [3]. Clinical data confirms this: nearly 59% of pilgrims are classified as high risk due to underlying conditions such as hypertension, diabetes, or advanced age, with an overall hospitalization rate of 971 per 100,000 pilgrims [4]. We believe customers will choose **Yusr** because our market research revealed consistent pain points across respondents.

Pilgrims frequently reported losing count of Tawaf and Sa'i rounds due to crowd distractions, often repeating extra rounds to be safe, creating unnecessary stress during worship. Separation anxiety was a recurring concern, particularly among those traveling with children or elderly companions, with reunion times often exceeding 30 minutes. Interviewees also highlighted that phones become unreliable in crowded and noisy environments, further validating the need for a wearable-based solution. These findings highlight the need for a reliable, hands-free wearable that addresses both safety and spiritual accuracy simultaneously.

Yusr customers will use the wristband to gain peace of mind by eliminating the anxiety of ritual miscounting through motion sensor-based automatic tracking, allowing them to dedicate their full attention to worship without manual interruption. The real-time health monitoring and SOS alert features ensure that pilgrims — especially the elderly or those with medical conditions — can receive timely assistance in emergencies. For vulnerable pilgrims, wearable devices that transmit real-time data to medical teams enable swift emergency response, and integration of real-time crowd alerts and smart navigation further reduces physical strain [5]. Real-time location sharing through the companion mobile app allows families and group leaders to remain connected even in the most crowded environments and helps parents instantly locate children within large pilgrim groups. Multi-language support ensures that pilgrims from all backgrounds can fully benefit from all features. Several key trends support the growth of **Yusr's** target market. Saudi Arabia continues to expand its pilgrimage infrastructure under Vision 2030, with smart city initiatives creating a receptive environment for technology-integrated solutions. In Hajj 2025, pilgrims were issued wearable smart bands integrated with GPS, health trackers, and emergency alert functions, validating the market readiness for such solutions [6]. The number of pilgrims is growing rapidly, with 2024 witnessing a 101% increase in Umrah performers from 2022 [7]. From a regulatory perspective, Saudi Vision 2030's Pilgrim Experience Program has launched the "Smart Hajj" initiative, actively encouraging wearable health tracking solutions to support pilgrim welfare [8].

4.2 Market Sizing

Our market sizing analysis for **Yusr** in Saudi Arabia is based on a targeted evaluation of individual pilgrims performing Hajj and Umrah. On the B2C side, we have analysed the number of pilgrims who require a reliable smart wristband to enhance their safety, health monitoring, and spiritual experience during their pilgrimage journey. This analysis is structured into three key components (TAM, SAM, and SOM) each providing both the estimated number of target consumers and the expected revenue based on our projected product price.

4.2.1 Total Addressable Market (TAM)

The Total Addressable Market (TAM) describes the full scope of demand for **Yusr**'s solution. It represents the highest level of revenue the startup could potentially achieve within the Hajj and Umrah ecosystem if adopted by all pilgrims who could make use of the **Yusr** smart wristband. To ensure accuracy and realism, this analysis is based on the latest complete and officially published statistics by the General Authority for Statistics (GASTAT) of Saudi Arabia for the year 2024 [9], which reflect the actual size of the Hajj and Umrah market over a full year.

According to the 2024 official statistics:

- **Number of Hajj pilgrims (N_{Hajj})** = 1,833,164
- **Number of Umrah performers (N_{Umrah})** = 35,802,452

Yusr is offered to individual users through two acquisition models: **direct purchase** and **short-term rental**. The following sections estimate the TAM for each model separately.

- **Purchase Model:**

Since there is no product in the market that directly matches all the features of the **Yusr** smart wristband, the pricing benchmark was derived from comparable fitness bands and smart bands available on Amazon. A competitive pricing strategy was applied, defined as setting a product's price relative to competitors' pricing structures [10], based on an analysis of similar devices in terms of form factor and functionality.

To support this analysis, a set of comparable products was selected to provide a representative view of the current market:

- Amazfit Band 7 (246.59 SAR) [11]
- HONOR Watch 4 (307.80 SAR) [12]
- HUAWEI Band 11 Pro (249 SAR) [13]

This approach ensures that the pricing benchmark reflects the diversity of the market while remaining aligned with **Yusr**'s intended positioning.

To estimate a representative unit price for **Yusr**, the average of these prices was calculated as follows:

$$\text{Estimated Unit Price} = \frac{246.59 + 307.80 + 249}{3} = 267.8 \text{ SAR}$$

The TAM for the **purchase model** is therefore calculated as:

$$\begin{aligned} TAM_{Purchase} &= (N_{Hajj} + N_{Umrah}) \times 267.8 \\ TAM_{Purchase} &= (1,833,164 + 35,802,452) \times 267.8 \\ &= 37,635,616 \times 267.8 \\ &= 10,078,817,965 \text{ SAR} \end{aligned}$$

This value represents the **maximum potential annual revenue** under the assumption that all Hajj and Umrah pilgrims **purchase** one **Yusr** smart wristband.

- **Rental Model:**

In addition to direct purchase, **Yusr** is offered to pilgrims through a short-term rental model. The daily rental price is set at 15% of the product's retail price, in line with established rental industry practices. According to Rentrax (2024) [14], products that require regular maintenance and have shorter lifespans are typically rented at 15–25% of their retail price.

Since the wristband is a technology product characterized by rapid obsolescence, frequent charging requirements, and continuous operational use during pilgrimage, this rental benchmark is considered relevant to **Yusr**. Accordingly, the lower bound of this range (15%) was selected to maintain affordability for pilgrims in a price-sensitive, seasonal market, while also encouraging wider adoption.

$$\text{Estimated Daily Rental Price} = 0.15 \times 267.8 = 40.17 \text{ SAR}$$

The TAM for the **rental model** is therefore calculated as:

$$\begin{aligned} TAM_{Rental} &= (N_{Hajj} + N_{Umrah}) \times 40.17 \\ TAM_{Rental} &= (1,833,164 + 35,802,452) \times 40.17 \\ &= 37,635,616 \times 40.17 \\ &= 1,511,822,695 \text{ SAR} \end{aligned}$$

This value represents the **maximum potential annual revenue** under the assumption that each Hajj and Umrah pilgrim **rents** one **Yusr** smart wristband once during their trip.

- **Final TAM (Adoption Mix-Based Estimate) :**

After estimating the TAM separately for the purchase and rental models, a combined market size is derived based on expected user adoption behavior. Consumer trends indicate a clear shift toward rental over ownership. A U.S. Chamber of Commerce analysis reports that approximately **57.3%** of consumers are willing to rent rather than buy well-made and trendy products [15]. In the Hajj and Umrah context, many paid services are already provided through rental or short-term booking models (e.g., mobility carts) [16], and the temporary nature of pilgrimage visits further supports short-term, access-based usage rather than ownership.

Based on this evidence, we adopt the 57.3% rental share reported by the source [15], with the remaining 42.7% assumed to opt for purchase. Accordingly, the final TAM is calculated using a weighted adoption mix rather than directly summing both models:

$$\begin{aligned} TAM_{Final} &= 0.427 \times TAM_{Purchase} + 0.573 \times TAM_{Rental} \\ TAM_{Final} &= 0.427 \times 10,078,817,965 + 0.573 \times 1,511,822,695 \\ TAM_{Final} &= 5,169,929,675 \text{ SAR} \end{aligned}$$

This value represents the estimated **Total Addressable Market (TAM)** assuming full market coverage under the mixed adoption of purchase and rental models.

4.2.2 Serviceable Addressable Market (SAM)

The Serviceable Addressable Market (SAM) represents the portion of the Total Addressable Market (TAM) that can be realistically targeted and served. It provides a more practical estimate of revenue potential based on expected user adoption and market readiness.

Since there are no direct statistics on the adoption of the **Yusr** smart wristband specifically, this analysis relies on evidence from the broader wearable technology market, focusing only on the health-monitoring aspect, which is one of **Yusr**'s core functionalities and the closest comparable feature available in existing studies. Accordingly, we use health-oriented wearable adoption as a proxy rather than overall usage.

A cross-sectional consumer survey reports that approximately 69.5% of respondents are willing to adopt wearable devices for health-related use cases [17]. This figure reflects user openness to wearable technology in the health-monitoring context, which aligns closely with **Yusr**'s health and safety tracking functions. To avoid overestimation, this adoption rate is used as a proxy for the serviceable share of the market.

Accordingly, the SAM is calculated as a proportion of the TAM using the following formula:

$$SAM = TAM \times \text{Proportion of Market Served}$$

Using the final estimated TAM and the wearable adoption rate of 69.5% [17], the SAM is calculated as:

$$SAM = TAM_{Final} \times 0.695$$

$$SAM = 5,169,929,675 \times 0.695$$

$$SAM = 3,593,101,124 \text{ SAR}$$

This represents the estimated serviceable market size for individual pilgrims who are likely to adopt wearable-based solutions such as **Yusr**.

4.2.3 Serviceable Obtainable Market (SOM)

The Serviceable Obtainable Market (SOM) represents the portion of the SAM that **Yusr** can realistically capture. It is estimated by dividing the adjusted SAM by the total number of market participants, which is calculated as the number of competitors plus one (**Yusr**). This approach assumes an equal distribution of market share among all market participants.

$$SOM = \frac{SAM}{\text{Number of Competitors} + 1}$$

$$SOM = \frac{3,593,101,124}{5 + 1} = \frac{3,593,101,124}{6} = 598,850,187 \text{ SAR}$$

Therefore, the SOM for **Yusr** is approximately 598,850,187 SAR

4.3 Competitors

This section analyzes the key competitors in the Hajj and Umrah technology market, highlighting their main features and limitations to identify gaps that **Yusr** aims to address.

1- Pilgrim App



Figure 5: Pilgrim App Logo

Pilgrim App is a comprehensive mobile application designed as an all-in-one guide for Hajj and Umrah. It provides step-by-step ritual instructions, real-time guidance, and AI-powered assistance. While it offers a manual Tawaf counter, a busyness checker for crowd monitoring, and a Dua collection, it remains entirely phone-dependent with no wearable integration, no automatic counting, no health monitoring, and no emergency SOS capabilities[18].

2- Tawaf Counter



Figure 6: Tawaf Counter App Logo

Tawaf Counter is a premium mobile application specifically designed for tracking Tawaf and Sa'i rounds using GPS-based assistance. It supports nine languages and offers offline functionality, making it accessible to international pilgrims. However, its GPS-dependent counting suffers from accuracy issues in crowded environments, and it lacks any health monitoring, emergency features, or wearable integration [19].

3- Umrah GPS Guide



Figure 7: Umrah GPS Guide App Logo

Umrah GPS Guide is an application that combines GPS-based automatic counting with a complete Umrah guide, emphasizing privacy through local data storage. While it offers offline GPS maps and a ritual checklist, it shares the same fundamental limitations as other apps — phone dependency, GPS inaccuracy in dense crowds, and the complete absence of health monitoring or emergency SOS features [20].

4- Al-Mutawwif



Figure 8: Al-Mutawwif App Logo

Al-Mutawwif is a Saudi-developed application serving as a comprehensive Hajj companion, offering detailed ritual guides, real-time crowd status monitoring, Haram maps, and prayer times. Despite its locally relevant design, it provides no automatic counting, no health monitoring, and no emergency response system, and remains fully dependent on mobile phone access throughout rituals[21].

5- Nusuk



Figure 9: Nusuk App Logo

Nusuk is the official government application developed by the Ministry of Hajj and Umrah in collaboration with SDAIA. It serves as the primary platform for permits, bookings, interactive maps, and comprehensive pilgrimage services. As the most widely adopted platform in the ecosystem, Nusuk excels in administrative and logistical support, yet it offers no wearable device, no automatic ritual counting, no advanced health monitoring, and no emergency SOS functionality [22].

Addressing Competitive Threats

While existing competitors have established presence in the Hajj and Umrah technology market, none offer a complete wearable solution. **Yusr** addresses each competitive threat through its integrated, hands-free approach. App-based competitors such as Pilgrim App, Al-Mutawwif, and Nusuk rely entirely on mobile phone interaction, which is impractical during physically demanding and spiritually focused rituals — **Yusr** eliminates this dependency through a wristband device that operates independently. Competitors offering automatic counting such as Tawaf Counter and Umrah GPS Guide depend on GPS technology, which suffers from significant accuracy issues in the dense crowds of the holy sites — **Yusr** addresses this through motion sensor-based counting, which is unaffected by network congestion or GPS signal loss. While **Yusr** does incorporate GPS, it is used exclusively for location tracking and group connectivity, further enhanced by indoor beacons deployed across the holy sites to ensure precise positioning even in dense crowd environments. Most critically, health monitoring and emergency response features are entirely absent across all competitors — **Yusr** fills this gap through continuous vital sign monitoring, health alerts, and a dedicated SOS system that connects pilgrims to emergency responders instantly. By combining all these capabilities into a single wearable device, **Yusr** positions itself as the only comprehensive safety and worship solution in the market.

4.4 Competitive Advantages

In this section, we present **Yusr's** key competitive advantages by highlighting the unique value it delivers compared to existing alternatives in the Hajj and Umrah technology market.

Yusr is the only commercially available wearable solution purpose-built for individual Hajj and Umrah pilgrims, offering a comprehensive set of features that no existing competitor provides. As shown in [Table 2: Competitive Analysis Comparison] below, all current competitors are mobile-only applications, leaving critical gaps in safety, ritual accuracy, and hands-free accessibility that **Yusr** uniquely addresses.

Table 2: Competitive Analysis Comparison

Features	Wearable Device	Auto Ritual Counter	Health Monitoring	Emergency SOS	Voice Comm.	Dua Collection	Step-by-Step Guide	Multi-Language	Gate Status
Yusr	✓	✓	✓	✓	✓	✓	✓	✓	✓
Pilgrim App	✗	✗	✗	✗	✗	✓	✓	✗	✗
Tawaf Counter	✗	✓	✗	✗	✗	✓	✓	✓	✗
Umrah GPS Guide	✗	✓	✗	✗	✗	✗	✓	✓	✗
Al-Mutawwif	✗	✗	✗	✗	✗	✓	✓	✗	✗
Nusuk	✗	✗	✗	✗	✗	✓	✓	✓	✓

Legend: ✓ = Available ✗ = Not Available

Yusr's key competitive advantages are:

- **Only commercially available wearable for individual pilgrims:** Eliminates phone dependency during rituals, enabling a truly hands-free pilgrimage experience something no existing app-based competitor can deliver.
- **Reliable automatic ritual counting:** Motion sensor–based counting ensures accuracy in dense crowds, unlike GPS-dependent competitors that suffer signal loss in the holy sites, giving pilgrims confidence in their worship without manual interruption.
- **Integrated health and emergency system:** Continuous vital sign monitoring with automatic SOS alerts a safety capability entirely absent across all competitors directly addressing the fact that nearly 59% of pilgrims are classified as high health risk [4].

- **Indoor precision tracking:** Real-time location sharing enhanced by deployed beacons across the holy sites, enabling group connectivity even in the most crowded environments where phones and calls become unreliable.

These advantages are sustainable because replicating a purpose-built wearable for Hajj and Umrah requires significant hardware investment, supply chain development, and deep domain knowledge of pilgrimage operations — barriers that protect **Yusr's** market position over time. Furthermore, the beacon infrastructure deployed across the holy sites creates a technical foundation that becomes increasingly difficult for competitors to replicate as the system scales.

5 Marketing Strategy

In this section, we define marketing goals aligned with the SMART criteria (Specific, Measurable, Achievable, Relevant, and Time-bound) to measure success, focusing on website traffic, social media growth, and registrations. As this is an early-stage campaign with no historical data, targets were estimated based on benchmarks. For website traffic, research indicates that new websites typically receive between 50 to 500 monthly visitors in their first months [23], making our target of 150 unique visitors realistic and well within this range. For registrations, early access and waitlist landing pages typically achieve conversion rates ranging from 25% to 85% [24], meaning our target of 100 registrations out of 150 expected visitors, representing a 67% conversion rate, falls well within this benchmark. For social media growth, our target of 100 followers per platform across WhatsApp, Instagram, and X was set based on benchmarks, which indicate that gaining 100 new followers within a short campaign period is considered a realistic and achievable goal for new accounts [25].

Table 3: Yusr Marketing Objectives, KPIs, and Targets

NO.	Objectives	KPIs	Targets
1	Increase website traffic by reaching at least 150 unique visitors within the six-day campaign, starting from April 3rd to April 8th, 2026.	Measured by the number of unique website visitors, tracked using the Base44 analytics dashboard .	Achieve 150 unique visitors within 6 days .

2	Achieve at least 100 user registrations on the landing page within the six-day campaign, starting from April 3rd to April 8th, 2026.	Measured by the number of actual completed registrations tracked using the Base44 system database .	Obtain 100 user registrations within 6 days .
3	Increase the number of followers on the Instagram platform by at least 100 followers within the six-day campaign, starting from April 3rd to April 8th, 2026.	Measured by the number of new followers gained on Instagram, tracked using Instagram insights .	Gain 100 Instagram followers within 6 days .
4	Increase the number of followers on the X platform by at least 100 followers within the six-day campaign, starting from April 3rd to April 8th, 2026.	Measured by the number of new followers gained on X, tracked using X analytics .	Gain 100 X followers within 6 days .
5	Increase the number of followers on our WhatsApp channel by at least 100 followers within the six-day campaign, starting from April 3rd to April 8th, 2026.	Measured by the number of new followers gained on the WhatsApp channel .	Gain 100 WhatsApp channel followers within 6 days .

5.1 Marketing Campaign

Effective marketing campaign is the foundation of **Yusr's** growth strategy, designed with clear objectives, defined target audiences, and measurable outcomes to ensure maximum impact. Throughout the campaign, we shared content that highlighted **Yusr's** unique features to drive traffic to our landing page. This section covers three key components: running the marketing campaign, the marketing channels, and the best performing channel.

5.1.1 Run the Marketing Campaign

The campaign execution phase involves launching and managing all planned marketing activities across selected channels. This includes coordinating content publishing schedules and ensuring all touchpoints are aligned with the campaign's core message and timeline. The campaign adopted a content marketing approach, leveraging engaging and informative content across all platforms to attract and convert the target audience. The campaign ran for 6 days, from April 3 to April 8, 2026.

5.1.2 Marketing Channels

To strategically accomplish our marketing objectives and maximize outreach to our target demographic—Hajj and Umrah pilgrims seeking a seamless and safe spiritual journey—we deployed a multi-channel marketing approach. We selected WhatsApp, Instagram, and X as our core platforms. This ecosystem allows for a blend of direct community engagement, visual storytelling, and real-time broadcasting.

5.1.2.1 WhatsApp

We chose WhatsApp as one of the major channels to engage with the audience directly. Considering its popularity and the option for group chats, WhatsApp helped us have direct and more personal interaction with potential customers. Our first activity involved building a dedicated WhatsApp channel to share engaging updates, including a short promotional video, and a direct link to our landing page, as shown in Figure 10.

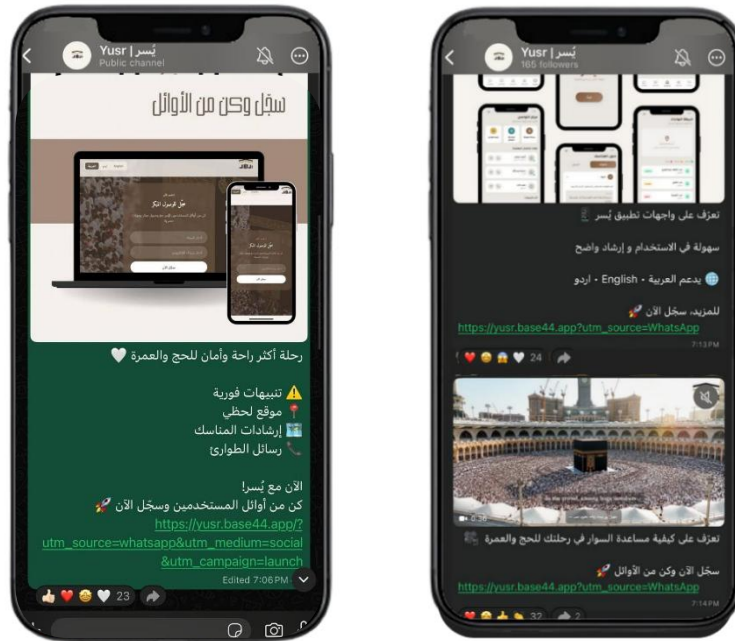


Figure 10: WhatsApp Marketing Channel

Additionally, we sent messages to multiple groups, providing a brief introduction to the **Yusr** wristband. As shown in Figure 11, the message included a welcoming introduction a short description of **Yusr's** key features—such as the Tawaf and Sa'i counter, real-time location tracking, emergency messages, ritual guides, and health sensors—and links to our landing page, WhatsApp channel, Instagram and X profiles. This strategy ensured that our audience was always informed and helped us reach a broad audience efficiently.



Figure 11: WhatsApp Marketing Message

- WhatsApp Channel URL: [Here](https://www.whatsapp.com/channel/0029Vb7AINXDTk1ymFhwCh3P)

5.1.2.2 Instagram

We used Instagram as a key platform to connect with our audience visually and build brand awareness. Since Instagram thrives on visual storytelling, we showcased **Yusr** through eye-catching images. On our Instagram bio, we included a brief description of **Yusr**, along with a trackable Base44 link to our landing page for easy access, as shown in Figure 12.



Figure 12: Instagram Marketing Profile

Our posts focused on highlighting the features and benefits of the **Yusr** wristband, building excitement for its release. To impress followers, we showcased visually appealing images of our uniquely designed smart wristband and how it provides a more comfortable and safe experience during Hajj and Umrah. Each post also included relevant hashtags to increase visibility and engagement. As shown in Figure 13.

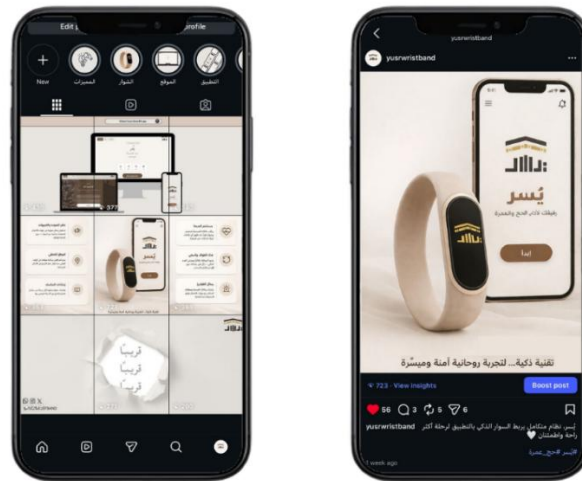


Figure 13: Instagram Marketing Posts

- Instagram URL: [Here](#)

5.1.2.3 X

Using X, we aimed to build an audience that would interact with **Yusr** and its features. X serves as an essential platform for sharing timely updates, including a short introductory video, and engaging with potential clients who might use the **Yusr** wristband. We improved our profile by adding a brief description of the wristband and providing our Base44 landing page link in the bio for direct access, as shown in Figure 14.



Figure 14: X Marketing Profile

We actively interacted with our audience by sharing several tweets that highlighted our competitive advantage, the safety and convenience features of the wristband, and our upcoming launch. Each tweet also included relevant hashtags to increase visibility and reach a wider audience, as shown in Figure 15.



Figure 15: X Marketing Posts

- X URL: [Here](#)

5.1.3 Best Channel

In this section, we will outline the channel that contributed the most traffic to our landing page. To identify the best channel for **Yusr**, we used Google's Campaign URL Builder to create custom tracking URLs integrated with our analytics tool, Base44. This allowed us to measure performance across channels and determine which platform generated the most traffic.

Based on analytical data retrieved from the Base44 platform during the campaign period as shown in Figure 11, **WhatsApp** was identified as the most effective marketing channel, based on the following factors:

- **Traffic Generation:** WhatsApp was the primary driver of traffic to the landing page, accounting for 58.5% of the total traffic with 196 unique visitors. This indicates the platform's high efficiency in reaching our target audience directly.
- **Engagement Quality:** The quality of engagement from WhatsApp was significantly higher, with an average session duration of 1 minute and 42 seconds. This suggests that users coming from this channel were more interested in exploring **Yusr's** features compared to other platforms.

In contrast, while platforms like Instagram and X contributed to the brand's visibility, they showed higher bounce rates and lower session durations. Therefore, WhatsApp is officially designated as the best channel for **Yusr**, proving to be the most reliable tool for driving high-quality engagement and sustainable user growth.

5.2 Campaign Performance

This section presents key analytics for our landing page, including total visits, unique visitors, average session duration, the percentage of traffic from search traffic, and visits via social referrals.

Our landing page was built using Base44 , which features a built-in analytics dashboard that enables direct performance monitoring. This dashboard offered comprehensive insights into our landing page's reach and effectiveness. Upon concluding the marketing campaign, we gathered several key metrics as outlined above, with Table 4 summarizing the overall marketing analysis.

Table 4: Overall Marketing Analysis

Number of visits	Number of unique visitors	Avg. session duration
764	335	19.6 $\approx$ 20 Seconds

The Base44 [26] analytics recorded 764 total visits from 335 unique visitors, with an average session duration of 19.6 seconds. All key metrics showed improvement compared to the previous period, reflecting sustained user interest alongside the overall traffic growth. The average session duration of 19.6 seconds reflects a reasonable level of engagement given the high number of visitors.

Figure 16 illustrates total visits to the landing page, highlighting two noticeable peaks throughout the period, with traffic remaining dynamic and active. Figure 17 presents unique visitors, showing a steadily increasing trend that peaks toward the end of the campaign period.

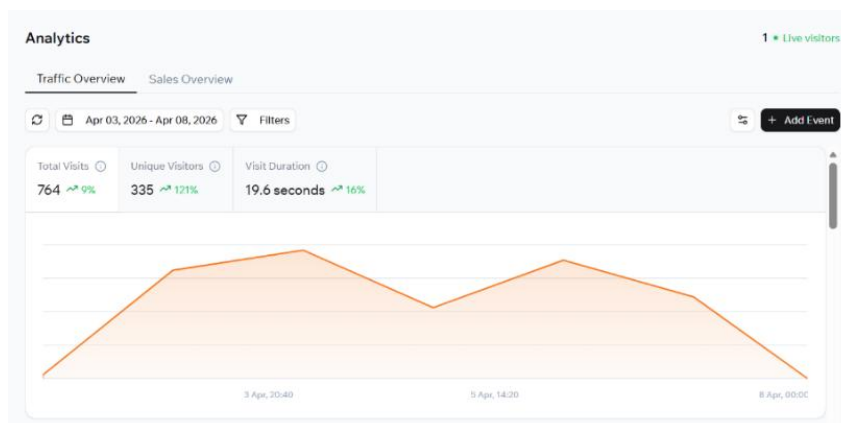


Figure 16: Total Visits to The Landing Page

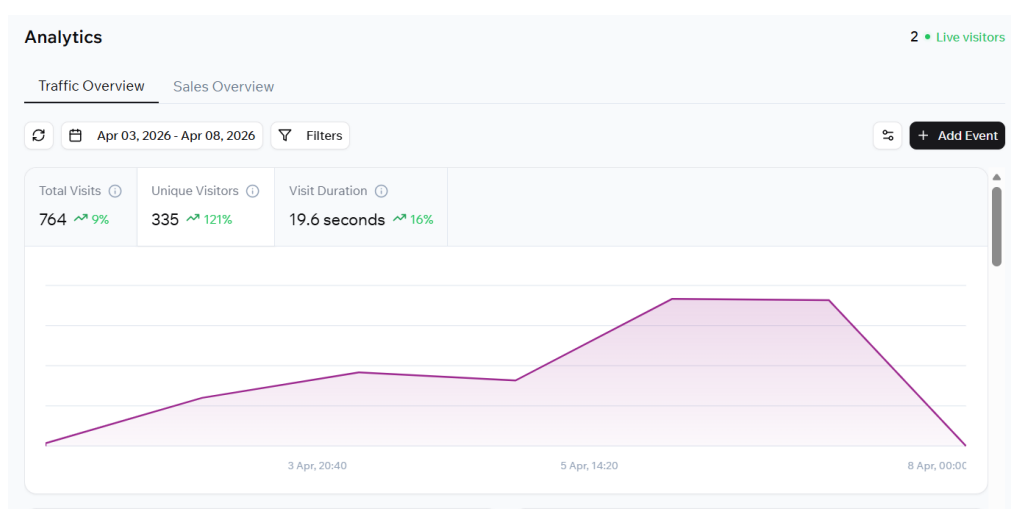


Figure 17: Number of Unique Visitors to The Landing Page

% Search Traffic

This section presents the percentage contribution of each traffic source to the total unique visitors, tracked through custom URLs generated via Google's Campaign URL Builder [27] as shown in Figure 20.

- **WhatsApp:** $\frac{196}{335} \times 100 \approx 58.5\%$
- **Instagram:** $\frac{77}{335} \times 100 \approx 22.9\%$
- **X:** $\frac{34}{335} \times 100 \approx 10.15\%$
- **Direct / None:** $\frac{28}{335} \times 100 \approx 8.36\%$

- The Total = 100%

Among all traffic sources, WhatsApp emerged as the dominant social referral channel as shown in Figure 20, significantly outperforming other platforms. This highlights its effectiveness in driving traffic, likely due to direct and personal sharing among users.

Regarding Direct / None traffic, it represented 8.36% of total visits, referring to visits that lack a clearly identifiable referral source. This includes users who accessed the landing page by typing the URL directly into their browser, clicking links embedded in desktop software or PDF documents, or accessing a bookmarked page [28].

Since Base44 does not provide search engine traffic analytics, Ubersuggest [29] was utilized to assess the landing page's organic search performance. As shown in Figure 18, the results confirm that the page does not rank for any organic keywords and recorded no search engine traffic. This aligns with our current strategy, as the campaign focused entirely on social media channels, resulting in limited visibility in search engines.

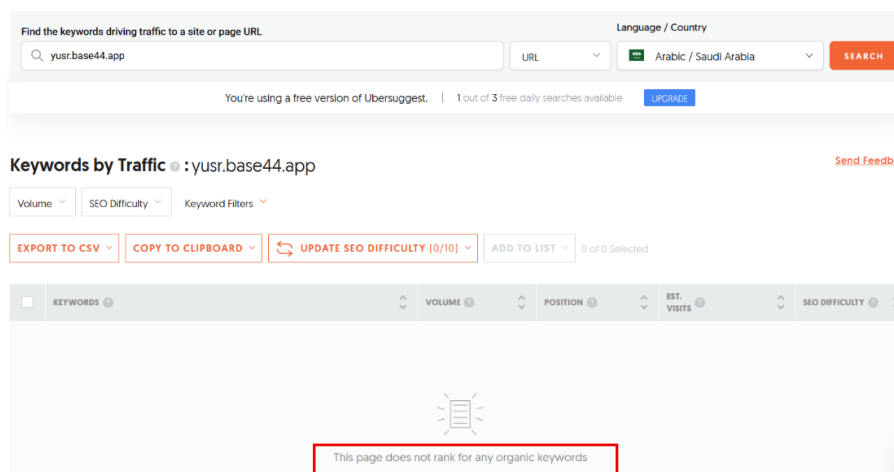


Figure 18: Search Results for Yusr

Geographical Distribution of Visits

To better understand the geographical distribution of visits to our landing page, which is crucial for assessing the effectiveness of our targeted efforts, Figure 19 shows that most visits originated from Saudi Arabia, accounting for 728 visits, representing 95.3% of total traffic, calculated as follows:

$$\frac{728}{764} \times 100 \approx 95.3\%$$

This aligns with our campaign's focus on the Saudi market. The remaining visits came from the United States (27), Jordan (5), Germany (1), Belgium (1), India (1), and Sweden (1), reflecting minimal but diverse international interest in the product.

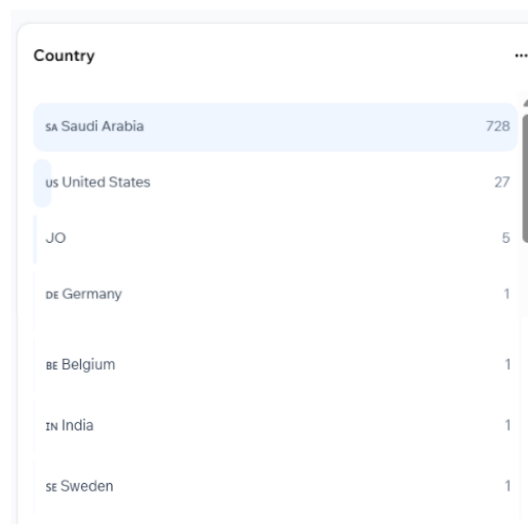


Figure 19: Geographical Distribution of Visits to Yusr's Landing Page

Number of Visits via Social Referral

In this section, we viewed the number of unique visitors referred from different sources from the Base44 dashboard as shown in Figure 11, summarized as follows:

- **WhatsApp: 196**
- **Instagram: 77**
- **X: 34**
- **Direct / None: 28**

The distribution of traffic sources indicates that WhatsApp was the predominant source, delivering 196 visitors and significantly outperforming other channels. This confirms its effectiveness in driving traffic, likely due to direct and personal sharing among users. Instagram and X contributed to a lesser extent, while Direct/None accounted for the smallest share of visits.

Visit Analytics

Top Sources

Source	Visitors	Bounce rate	Visit duration
WhatsApp	196	43%	1m 42s
Instagram	77	69%	1m 06s
X	34	85%	29s
Direct / None	28	54%	46s

Figure 20: Distribution of Traffic Sources

Figure 21 presents a sample of the visit log recorded during the campaign, displaying visitor details including source, medium, visit at, duration, and bounce. The complete dataset is provided in the

Excel file.

visitor_id	source	medium	campaign	visited_at	duration
560da92d-a6c8-402...	Instagram	social	launch	2026-04-03 18:24:19	5
6672df33-2cb2-46d...	WhatsApp	social	launch	2026-04-03 18:20:31	110
9b0ef54c-3ba3-46c...	Instagram	social	launch	2026-04-03 18:12:26	48
473f8511-0915-4204...	X	social	launch	2026-04-03 17:51:20	31
5f6457a8-4989-4c3...	X	social	launch	2026-04-03 17:42:30	7

Figure 21 : Sample of The Visit Log

5.3 Web Registration

This section presents the registration data collected through the **Yusr** landing page during the six-day campaign period. The landing page successfully collected **147 registrations**, where visitors submitted their full name and email address to register for early access. Out of **335 unique visitors**, **148 registered**, reflecting a conversion rate of approximately **44.2%**. This indicates that nearly half of all unique visitors were sufficiently interested in **Yusr** to sign up, demonstrating strong audience intent and effective messaging across all channels. As shown in Figure 22, the registration data was exported and documented as evidence of campaign results. The complete dataset is provided in the Excel file.

full_name	email	registered_at
حنينه ربيع	biborabie@gmail.com	2026-04-08T12:28:46.480Z
أريج	ayereej@hotmail.com	2026-04-08T11:59:50.451Z
ساره	s0.sarona.sa@gmail.com	2026-04-08T10:00:21.134Z
ريان بلال الشمري	ryanbelal200@gmail.com	2026-04-07T23:57:29.665Z
Naser	zflir9@gmail.com	2026-04-07T20:21:37.539Z
Mahmoud	mahmoud77saif@gmail.com	2026-04-07T20:21:06.195Z

Figure 22:Yusr Beta Registration Database

5.4 Overall Campaign Results

To evaluate the effectiveness of the six-day marketing campaign, performance was measured against predefined objectives and key performance indicators (KPIs). The following sections provide a structured, data-driven analysis of campaign outcomes across all digital channels, covering the comparison between target and actual results, overall campaign success, and key lessons learned.

5.4.1 Assessing Campaign Effectiveness

The **Yusr** marketing campaign was designed around five SMART-aligned objectives targeting user acquisition, engagement, and conversion across multiple digital platforms, including WhatsApp, Instagram, and X. Throughout the campaign period, performance was continuously monitored using the Base44 analytics dashboard alongside platform-specific insights to ensure accurate evaluation against KPIs as shown in Table 1.

Overall, the campaign demonstrated a high level of effectiveness, with all predefined objectives successfully achieved and exceeded. Landing page registrations reached 148 against a target of 100, while unique visitors reached 335 compared to the target of 150. Similarly, Instagram achieved 238 followers, X reached 137 followers, and WhatsApp reached 165

followers, all exceeding their respective targets as shown in Figure 23. These results indicate a strong alignment between the campaign strategy and audience behavior.

The integrated multi-channel strategy combining direct WhatsApp communication, visually engaging Instagram content, and real-time updates on X played a key role in driving awareness and engagement for **Yusr's** smart wristband designed for Hajj and Umrah pilgrims.



Figure 23: Yusr's Followers on Instagram (238), WhatsApp Channel (165) and X (137)

5.4.2 Comparison of Actual Results vs Targets

This section presents a detailed side-by-side comparison between the pre-defined targets and the actual results. As shown in Table 5, all five objectives were not only met but exceeded, reflecting the strong reach and engagement generated by the campaign.

Table 5: Yusr Campaign KPI Targets vs. Actual Results

Objective	KPI	Target	Actual Result	Status
Achieve 150 unique visitors within 6 days	Number of unique visitors (Base44)	150 unique visitors	335 unique visitors	Exceeded
Obtain 100 user registrations within 6 days	Number of registrations (Base44)	100 registrations	148 registrations	Exceeded
Gain 100 Instagram followers within 6 days	New followers on Instagram	100 followers	238 followers	Exceeded
Gain 100 X followers within 6 days	New followers on X	100 followers	137 followers	Exceeded
Gain 100 WhatsApp followers within 6 days	New followers on WhatsApp channel	100 followers	165 followers	Exceeded

5.4.2 Campaign Success Evaluation

Overall, the **Yusr** marketing campaign was highly successful, achieving and surpassing all predefined KPIs within the six-day period. The findings from the Base44 analytics dashboard and social media platforms confirm the effectiveness of the multi-channel strategy in reaching and engaging the target audience.

- **Landing Page:** By utilizing the analytics dashboard, we tracked our performance and recorded 764 total visits with 335 unique visitors and an average session duration of 19.6 seconds. As shown in Figure 24, the dashboard provides a comprehensive overview of the landing page performance throughout the campaign period.

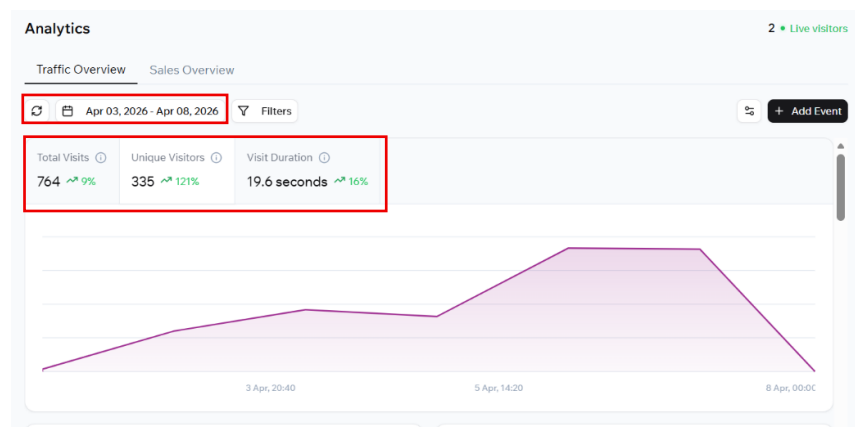


Figure 24: Base44 Analytics Dashboard Overview

The landing page successfully collected 148 registrations, achieving a conversion rate of 44.2% based on total unique visitors, as detailed in [Section 5.3].

- **Social Media Channels:** The campaign was activated across three platforms: Instagram, WhatsApp, and X, each contributing to building awareness and driving traffic to the landing page.

1) WhatsApp: WhatsApp proved to be the dominant traffic driver, delivering 196 unique visitors to the landing page accounting for 58.5% of total traffic, with the longest average visit duration of 1 minute and 42 seconds (See Figure 20), indicating highly engaged visitors. This strong performance is attributed to the effectiveness of direct messaging platforms in reaching families planning Hajj and Umrah, where content shared through community channels tends to spread organically through group sharing and forwarding.

As shown in Figure 25, our most engaging post on the WhatsApp channel was a promotional video about Yusr, presenting its value in supporting pilgrims during their journey, along with brief caption and a direct registration link. The post received 32 reactions from followers, reflecting strong audience interest and engagement with the product.

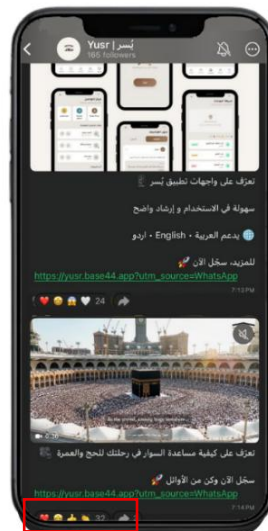


Figure 25: WhatsApp Channel — Most Engaging Post

As previously noted in [Section 7.1], the WhatsApp channel successfully exceeded its target by the end of the campaign, as illustrated in Figure 26.

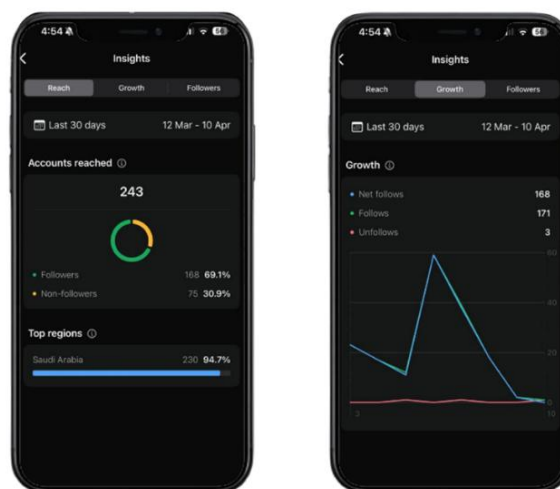


Figure 26: WhatsApp Channel Reach & Growth Analysis

Note: WhatsApp analytics does not support custom date filtering for the campaign period (April 3–8). The figures above reflect the last 30 days (March 12 – April 10).

2) Instagram: Instagram contributed 77 unique visitors to the landing page, accounting for 22.9% of total traffic, with an average visit duration of 1 minute and 6 seconds as shown in Figure 20, indicating meaningful engagement with our content.

As previously mentioned in [Section 5.3], Instagram ranked second in landing page traffic, and as previously mentioned in [Section 7.1] Instagram ranked first among all channels in follower growth, reaching 238 followers by the end of the campaign as illustrated in Figure 27.

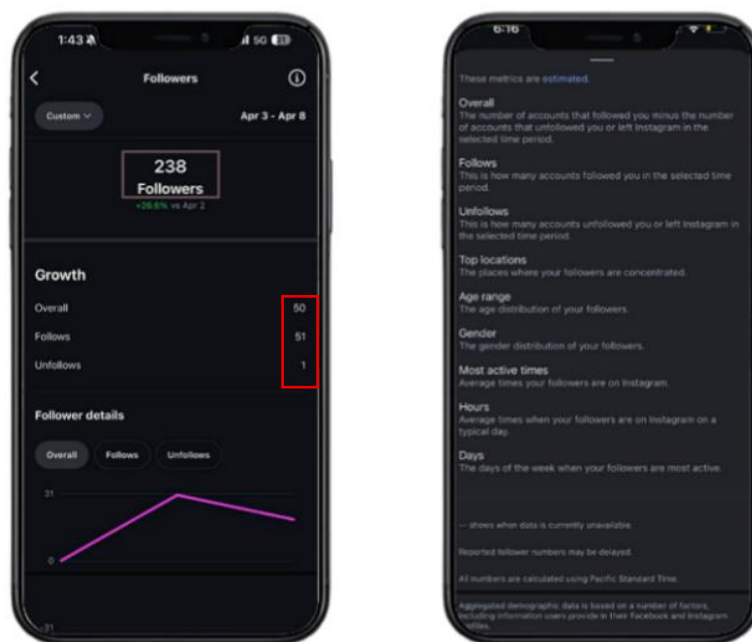


Figure 27: Instagram Followers Growth Analysis

Note: As indicated by Instagram, reported follower numbers may be delayed. The follower count of 238 reflects the actual count at the end of the campaign on April 8, as the insights dashboard may not immediately reflect real-time changes.

Our most engaging post, featuring a product photo of the **Yusr** wristband published on April 3, was the top-performing content across the campaign. As shown in Figure 28, the post generated 703 total views with 55.5% of views coming from non-followers, demonstrating strong organic reach beyond our existing audience.

The post recorded 70 total interactions: 55 likes, 3 comments, 6 shares, 5 reposts, and 1 save. It also drove significant profile activity, including 44 profile visits, 11 external link taps to the landing page, and 9 direct new follows directly attributed to this post alone.

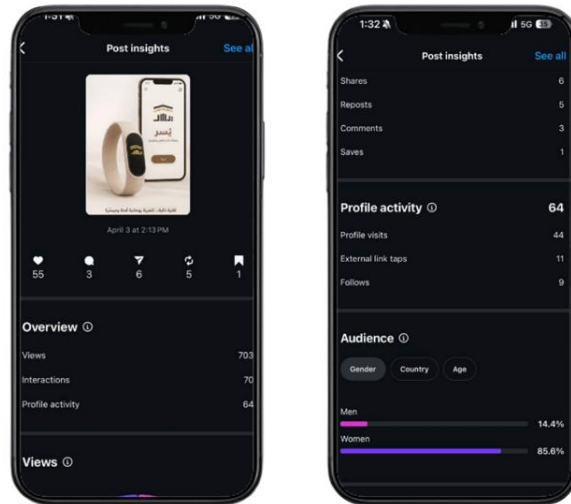


Figure 28: Instagram Most Engaging Post Overview

Figure 29 provides a detailed breakdown of the post's interaction metrics. Notably, 87.1% of interactions came from existing followers, while 12.9% came from non-followers, demonstrating the post's success in reaching new audiences.



Figure 29: Most Engaging Post — Interaction Breakdown

Figure 30 shows the post's views analysis. Of the 703 total views, the top sources were Profile (35.7%), Stories (31.9%), and Feed (22.2%), with 55.5% of views coming from non-followers confirming the post's ability to attract new audience members organically. The post reached 319 accounts in total.



Figure 30: Most Engaging Post — Views & Reach Analysis

At the account level, (Figure 31) shows the overall views performance during the campaign period (April 3–8). The account generated 4,049 total views, with 58.7% from followers and 41.3% from non-followers, reaching 386 accounts — a +19,200% increase compared to the prior period (March 28 – April 2). Content was distributed across Posts (76.3%) and Stories (23.7%).

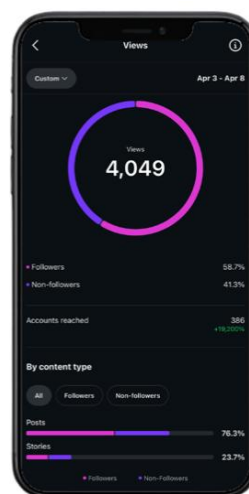


Figure 31: Instagram Account Views & Reach Overview

Figure 32 illustrates the account's profile activity during the campaign period. The account recorded 1,106 total profile activity actions — a +8,407.7% increase compared to the prior period (March 28 – April 2). This included 1,040 profile visits (+8,566.7%) and 66 external link taps (+6,500%), confirming that our Instagram content was highly effective in driving users to visit our profile and click through to the landing page. The top performing posts by views were published on April 3, with 703, 412, 371, and 358 views respectively.

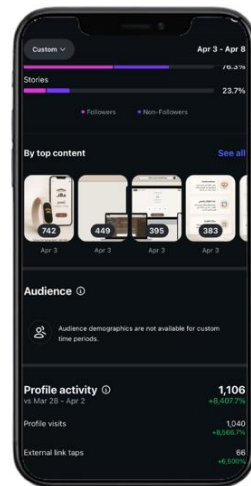


Figure 32: Instagram Profile Activity & Top Content Views

Figure 33 presents the overall interaction summary for the campaign period, with 439 total likes, 22 comments, 29 shares, 31 reports, and 1 save across all posts. The top posts based on likes were led by the landing page post (57 likes), followed by the product photo (55 likes), confirming that product-focused visual content resonated most strongly with our audience.

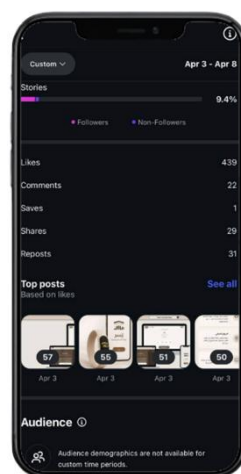


Figure 33: Instagram Overall Campaign Interactions & Top Posts

Finally, Figure 34 presents the audience demographics of our Instagram followers. Our audience was predominantly female (94.8% Women, 5.2% Men) and concentrated in the 18–24 age group (51.7%), followed by 25–34 (29.3%) and 35–44 (13.8%). Geographically, 84.1% of followers are in Riyadh, with additional followers in Dammam (3.8%), Ad Dam, Ar Riyad (3.0%), Hafar Al Batin (3.0%), and Amman (1.5%). This demographic profile closely aligns with our campaign's focus on the Saudi market.

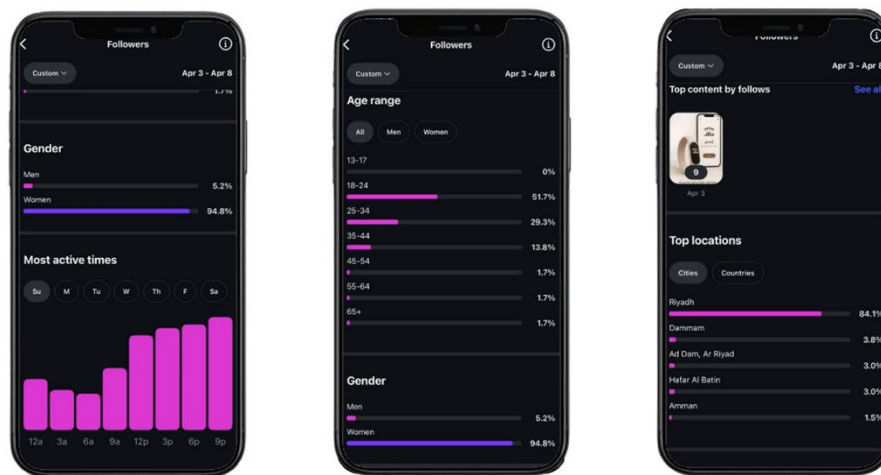


Figure 34: Instagram Audience Demographics — Age, Gender & Top Locations

3) X: X contributed 34 unique visitors to the landing page (10.15% of total traffic). Despite having the highest bounce rate at 85% (See Figure 20), the platform generated strong engagement metrics across posts. This suggests that while users interacted with the content on X, fewer users were motivated to continue to the landing page, resulting in lower conversion performance compared to other channels. Two posts in particular stood out as the top performers during the campaign period, as detailed in the following figures.

As shown in Figure 35, our most engaging post a landing page registration call-to-action published on April 3 with the hashtags — **#حج_عمرة_تقنية** — generated 2,290 impressions, 146 total engagements, and 71 profile visits. The post received 26 likes and 3 reposts, with 3 detail expands, reflecting strong audience curiosity. These results indicate that the post was effective in capturing attention and encouraging users to visit the profile, although the conversion to landing page visits remained limited compared to other platforms.



Figure 35: Most Engaging Post Activity

Figure 36 presents our second top-performing post a 39-second introductory video of Yusr published on April 3 with the caption — “رحلة أسهل تبدأ مع يُسر” — which generated 1,244 impressions, 71 engagements, and 30 profile visits. The video received 28 likes, 2 reposts, and 2 replies, with 5 detailed expands. The video section recorded 301 total views with 165 unique viewers. The audience retention graph shows highest engagement in the first 9 seconds, which is typical for short-form video content and confirms that the opening of the video effectively captured viewer attention.

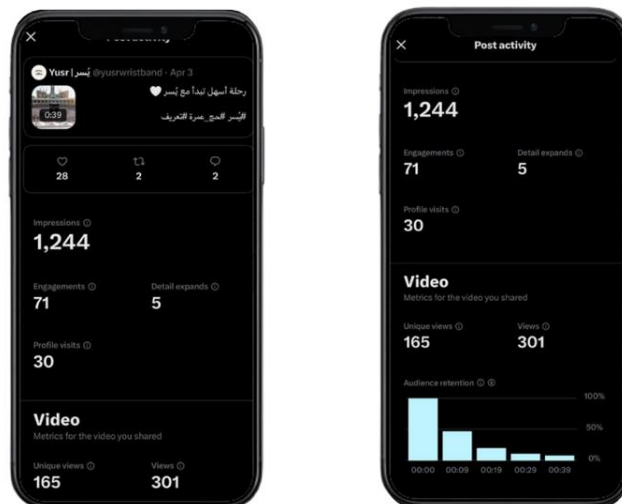


Figure 36: Video Post Activity & Audience Retention

Figure 37 shows the overall X account analytics during the campaign period (April 3–8). The account generated approximately 5,300 total impressions with an engagement rate of 11.1%, recording 589 total engagements, 197 likes, 17 reposts, 5 replies, and 4 bookmarks.

Profile visits reached 109, reflecting strong audience curiosity driven by our content. The follower growth graph shows consistent daily growth throughout the campaign, with notable spikes on April 6 and 7, confirming that our content effectively attracted new followers over the campaign period.

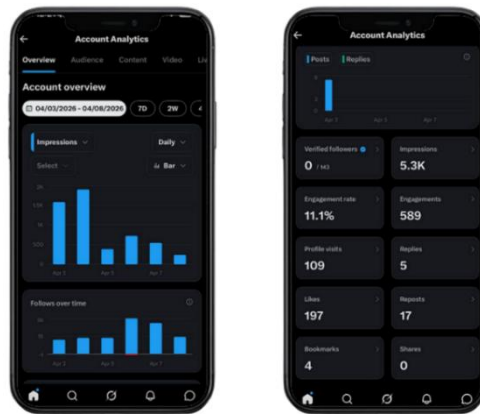


Figure 37: Account Analytics & Follower Growth

Note: The verified followers count shown in the analytics (143) does not support custom date filtering and reflects a broader period. The actual follower count at the end of the campaign on April 8 was 137, as confirmed from the profile.

Figure 38 presents the X audience demographics during the campaign period (April 3–8). The results indicate that the audience was predominantly based in Saudi Arabia, with the 18–24 age group representing the highest level of engagement, followed by both the 25–34 and 13–17 age groups equally in second place. In terms of gender distribution, the audience was predominantly female, reflecting the engagement patterns observed during the campaign period.



Figure 38: X Platform — Audience Demographics

The distribution of traffic sources across all channels is presented in [Section 5.2], which provides a comprehensive analysis of visitor referrals, bounce rates, and session durations, supporting the evaluation of channel effectiveness.

5.5 Lessons Learned

This section presents key insights, challenges, and strategic recommendations derived from the campaign performance across all platforms.

1. Insights

- WhatsApp was the most effective channel for driving traffic, accounting for 58.5% of all landing page visits. This demonstrates the effectiveness of direct messaging platforms in reaching families planning Hajj and Umrah, where content shared through community channels tends to spread organically through group sharing and forwarding.
- Instagram delivered the second-highest traffic volume (22.9%) with a strong average visit duration of 1 minute and 6 seconds, suggesting that Instagram's highly visual feed format and product-focused imagery effectively captured audience interest and motivated users to explore Yusr's offering further.
- The conversion rate of 44.2% (148 registrations from 335 unique visitors) indicates that the landing page content was compelling and well-aligned with the needs and expectations of our target audience.
- Highlighting the landing page's multilingual support (Arabic, English, and Urdu) in our social media posts may have contributed to attracting international users, as reflected in visits from outside Saudi Arabia, suggesting growing global interest in the product.

2. Challenges

- Despite the overall success of the campaign, several limitations were identified. X recorded the highest bounce rate at 85%, along with the shortest average session duration of 29 seconds. This indicates a weaker alignment between platform audience behavior and landing page content, resulting in lower retention compared to other channels.

- The six-day campaign duration limited long-term organic growth, particularly on platforms such as X, which require longer periods to build consistent engagement.

3. Recommendations for the Next Campaign

- Based on the performance observed, several strategic improvements are recommended for future campaigns. WhatsApp should remain the primary communication channel due to its proven effectiveness in driving high-quality traffic and engagement. Expanding broadcast lists and leveraging community-based distribution networks could further enhance reach.
- Instagram content strategy should place greater emphasis on Reels and Stories, as these formats are more likely to achieve higher organic reach and sustained engagement compared to static posts.
- For X, a revised content strategy is required to improve engagement quality and reduce bounce rates. This may include the use of interactive formats such as polls, discussion threads, and trending, context-specific hashtags to better align with platform behavior.
- Extending the campaign duration to a minimum of two weeks is also recommended to allow for more reliable data collection, improved optimization cycles, and stronger organic growth across all platforms.
- Finally, incorporating influencer collaborations within the Hajj and Umrah niche is expected to enhance credibility, broaden audience reach, and strengthen trust in the **Yusr** brand proposition.

6 Financial Analysis

6.1 Considered Costs

This section provides a comprehensive breakdown of all costs associated with **Yusr**'s first year of operations. Costs are classified into two categories: fixed costs, which remain constant regardless of production volume, and variable costs, which fluctuate in proportion to production activity and sales.

6.1.1 Fixed costs

Fixed costs are expenses that remain constant regardless of production volume or sales. These costs must be covered regardless of revenue levels, making accurate estimation critical for financial sustainability planning.

6.1.1.1 Employee Salaries

Yusr's team consists of twelve employees across eight roles, hired at market rates as referenced from GulfTalent Saudi Arabia [30][35][37], Glassdoor [31], Paylab Saudi Arabia [34], and WorldSalaries [32][33][36]. As a startup, **Yusr** adopts a lean staffing model with one employee per role to minimize overhead costs while maintaining full operational coverage. An exception is made for the Sales Employee role, where five employees are assigned — one per distribution booth at Al-Masjid Al-Haram — as physical presence at each booth is essential for direct customer engagement during pilgrimage seasons. The total monthly salaries cost is $\approx 103,980$ SAR, resulting in a total annual salaries cost of $\approx 1,247,776$ SAR, as detailed in [Table 6 :Employee Salaries].

6.1.1.2 App & Website Development

The **Yusr** mobile application is classified as a complex app given its integration with IoT hardware (wristband), real-time GPS tracking, multi-language support, and health data analytics. Based on TrangoTech's app development cost guide for the Saudi market [38], complex app development costs range between 450,000 and 750,000 SAR, and AI app development between 200,000 and 650,000 SAR.

Since **Yusr** incorporates advanced AI features within a complex application, the upper range of the combined categories is considered, yielding an estimated development cost of:

$$(200,000 + 750,000) \div 2 = 475,000 \text{ SAR}$$

Publishing on the Apple App Store requires an annual fee of \$99 (≈ 371 SAR) [39], while the Google Play Store charges a one-time fee of \$25 (≈ 94 SAR) [40]. Annual app maintenance averages 83,125 SAR/year based on 15–20% of development cost [41]. Website development averages 30,000 SAR based on multiple sources [42][43], with annual maintenance averaging 5,250 SAR/year. The full breakdown is shown in [Table 7: App & Website Development]

6.1.1.3 Cloud Services & Hosting

Yusr will utilize Amazon Web Services (AWS) hosted in the Middle East (Bahrain) region. The infrastructure includes Amazon EC2 for application hosting (\$48.91/month), Amazon RDS for the managed database (\$165.44/month), and Amazon S3 for data storage (\$0.23/month). Based on the AWS Pricing Calculator [44], the total is approximately 9,655.35 SAR/year. The full breakdown is shown in [Table 8 : Cloud Services & Hosting].

6.1.1.4 Marketing & Advertising

Yusr employs a targeted multi-channel marketing strategy centered around the three key pilgrimage seasons: the Hajj season, Ramadan (peak Umrah period), and the general Umrah period in January. Three marketing campaigns are launched annually, timed approximately three months before each season to capture pilgrims during their planning and booking phase. The marketing mix consists of six channels: Instagram, X, TikTok, YouTube, Search Engine Optimization (SEO), and Pilgrimage Platform Advertising.

To balance reach and trust, **Yusr** employs a two-tier influencer strategy across Instagram, TikTok, and YouTube. Micro-influencers (10K–100K followers) are leveraged for authentic engagement and community trust, as pilgrims tend to relate more to peer-level voices when making faith-based purchasing decisions. Macro-influencers (500K–1M followers) are engaged to drive mass brand awareness among both domestic and international pilgrim populations.

- **Instagram:** Instagram is leveraged as a primary visual channel to showcase **Yusr's** wristband features through demonstration videos and infographics targeting Hajj and Umrah pilgrims. Based on market benchmarks for startups in Saudi Arabia, the average monthly Instagram advertising spend ranges between SAR 1,000 and SAR 3,000 [45]. Adopting the midpoint of SAR 2,000 per month, and running three one-month campaigns per year, the total annual Instagram advertising cost is:

$$SAR\ 2,000/month \times 3\ campaigns = 6,000\ SAR/year$$

In addition, two micro-influencers and two macro-influencers are engaged per campaign across three annual campaigns. Based on industry benchmarks [46], micro-influencer Instagram

posts are priced at approximately \$300 ($\approx$ SAR 1,125) per post. Macro-influencer rates are approximately \$7,500 per post [46][47]:

- Micro: $\$300/\text{post} \times 2 \text{ influencers} \times 3 \text{ campaigns} = \$1,800 \times 3.75 = 6,750 \text{ SAR/year}$
- Macro: $\$7,500/\text{post} \times 2 \text{ influencers} \times 3 \text{ campaigns} = \$45,000 \times 3.75 = 168,750 \text{ SAR/year}$

Total Instagram: SAR 181,500/year

- **X:** is particularly popular in Saudi Arabia for real-time news and trend engagement, making it an effective channel for reaching pilgrims during key planning moments. According to Circleboom (2025), small businesses typically allocate \$30–\$50 per day for X advertising [48]. Applying a daily budget of \$40 (approximately SAR 150) over 30 days per campaign, and running three campaigns per year, the total annual X advertising cost is:

SAR 4,500/month $\times$ 3 campaigns = 13,500 SAR/year

- **TikTok:** TikTok is utilized through a two-tier influencer approach targeting pilgrims across domestic and international markets. Micro-influencers (10K–100K followers) deliver higher engagement and authenticity, while macro-influencers (500K–1M followers) maximize reach and brand awareness. Based on Business of Apps (2024) benchmarks, micro-influencer TikTok posts are priced at approximately SAR 422 per post [49], and macro-influencer posts at approximately \$10,000 per post [46]. Two micro-influencers and two macro-influencers are engaged per campaign across three annual campaigns:

- Micro: $\text{SAR } 422/\text{post} \times 2 \text{ influencers} \times 3 \text{ campaigns} = 2,532 \text{ SAR/year}$
- Macro: $\$10,000/\text{post} \times 2 \text{ influencers} \times 3 \text{ campaigns} = \$60,000 \times 3.75 = 225,000 \text{ SAR/year}$

Total TikTok: 227,532 SAR/year

- **YouTube:** is used to publish demonstration videos and product explainers targeting international pilgrims researching **Yusr** prior to their journey. Saudi Arabia's YouTube CPM ranges between \$8 and \$14 per thousand impressions for technology and consumer content [50]. A monthly campaign budget of \$300 (approximately SAR 1,125) is allocated per campaign, running three campaigns annually:

$$1,125 \text{ SAR /month} \times 3 \text{ campaigns} = 3,375 \text{ SAR/year}$$

In addition, two micro-influencers and two macro-influencers are engaged per campaign across three annual campaigns. Based on industry benchmarks, micro-influencer YouTube sponsored integrations are priced at approximately \$500 ($\approx$ SAR 1,875) per post [46], and macro-influencer posts at approximately \$15,000 per post [49]:

- Micro: $\$500/\text{post} \times 2 \text{ influencers} \times 3 \text{ campaigns} = \$3,000 \times 3.75 = 11,250 \text{ SAR/year}$
- Macro: $\$15,000/\text{post} \times 2 \text{ influencers} \times 3 \text{ campaigns} = \$90,000 \times 3.75 = 337,500 \text{ SAR/year}$

Total YouTube: 352,125 SAR/year

- **Search Engine Optimization (SEO):** SEO is applied to enhance the visibility of Yusr's official website in search engine results, enabling pilgrims to discover the product during their Hajj and Umrah planning phase. SEO targets bilingual (Arabic, English and Urdu) keywords related to pilgrimage safety, smart wristbands, and ritual tracking. Unlike seasonal ad campaigns, SEO is maintained year-round to build sustained organic traffic. Based on a pricing guide for SEO services in Saudi Arabia [51], the monthly cost for SEO services is 10,000 SAR, resulting in an annual SEO cost of:

$$10,000 \text{ SAR /month} \times 12 \text{ months} = 120,000 \text{ SAR/year}$$

- **Pilgrimage Platform Advertising:** Unlike seasonal campaigns, Pilgrimage Platform Advertising is maintained year-round to capture pilgrims planning their journey at any time. Based on regional digital travel platform advertising benchmarks [52], a budget of SAR 3,000 is allocated per month:

$$3,000 \text{ SAR/month} \times 12 \text{ months} = 36,000 \text{ SAR/year}$$

- **Marketing Agency Fee:** It is important to note that the advertising costs listed above (Instagram, X, TikTok, YouTube, SEO, and Pilgrimage Platform) represent direct platform spend the amounts paid directly to each advertising platform. These are separate from, and in addition to, the cost of hiring a professional marketing agency to plan, execute, and manage all campaigns across all channels. To ensure consistent brand messaging, bilingual

content creation (Arabic, English and Urdu), and expert campaign optimization year-round, **Yusr** engages a full-service digital marketing agency.

Based on market pricing for comprehensive multi-platform marketing packages in Saudi Arabia [53], monthly retainer fees for agencies managing four or more platforms with full campaign management range from SAR 10,000 to SAR 30,000. Adopting SAR 25,000 per month as a representative rate for a specialized agency managing **Yusr's** multi-channel pilgrimage marketing strategy:

$$25,000 \text{ SAR /month} \times 12 \text{ months} = 300,000 \text{ SAR/year}$$

The full breakdown is shown in [Table 9 : Marketing & Advertising].

6.1.1.5 Hardware & Software Tools

To support daily operations, the following hardware and software tools will be procured, as detailed in [Table 10 : Hardware & Software Tools]:

- **Laptops:** Each of the six employees in addition to the project manager will be equipped with a MacBook Pro 16" M4 laptop (8,499 SAR each) [54], totalling 59,493 SAR as a one-time payment with an approximate lifespan of five years.
- **iPhone:** One iPhone (7,299 SAR each) [55] will be purchased for IOS application testing, totaling 7,299 SAR as a one-time payment.
- **Android Phone:** One android phone (5,499 SAR each) [56] will be purchased for Android application testing, totaling 5,499 SAR as a one-time payment.
- **iPads:** 5 iPads (5,499 SAR each) [57] will be purchased for Services Booths, totaling 27,495 SAR as a one-time payment.
- **Microsoft 365 Business Basic:** Seven subscriptions at \$12.50 ($\approx$ 47 SAR) per user per month [58], totaling $\approx$ 3,938 SAR/year. Includes Word, Excel, PowerPoint, OneDrive, Teams, and Outlook.
- **Office Printer (HP LaserJet):** One printer at 2,299 SAR [59] as a one-time payment for administrative documentation, with an approximate lifespan of five years.

6.1.1.6 Beacons Setup

- **Hardware Cost (Kontakt.io Anchor Beacon 2 [60]):** Using \$35/unit $\approx$ 131.25 SAR/unit as a conservative enterprise bulk price benchmark, cited from Kontakt.io's own pricing context and the eBay secondary market floor:

$$2,000 \times 131.25 \text{ SAR} = 262,500 \text{ SAR (one-time)}$$

- **Installation Labor :** Kontakt.io states that beacon installation normally costs around \$100 per item[61], covering the salaries and expenses of the installation team. This includes the specialized IoT technicians who mount, configure, test, and register each beacon into the cloud management system. Applying \$100/unit $\approx$ 375 SAR for the full installation service:

$$2,000 \times 375 \text{ SAR} = 750,000 \text{ SAR (one-time)}$$

- **Annual Maintenance:** $262,500 \times 10\% = 26,250 \text{ SAR/year}$
- **Kontakt.io Cloud Platform (Kio Cloud):** Enterprise beacon management platform — based on Kontakt.io's SaaS pricing model for 2,000 devices, estimated as follow:

$$2,000 \text{ SAR/month} \times 12 = 24,000 \text{ SAR/year}$$

The full breakdown is shown in [Table 11 : Beacons Setup Costs]

6.1.1.7 Cybersecurity & Data Protection Compliance

As **Yusr** collects sensitive personal data — including heart rate metrics, GPS location, and personal identifiers from pilgrims across 184 countries, compliance with Saudi Arabia's Personal Data Protection Law (PDPL) and international information security standards is mandatory. The PDPL, fully enforced since September 2024, imposes fines of up to SAR 5,000,000 for non-compliance [62]. **Yusr** will pursue ISO 27001 certification the globally recognized information security management standard to demonstrate compliance and protect pilgrim data.

Based on industry benchmarks for small organizations under 50 employees [63], the Year 1 cost is calculated as follows:

- **ISO 27001 Implementation & Certification (Y1):** First-time ISO 27001 certification for small organizations costs between \$15,000 and \$50,000 [63]. Adopting the upper range given **Yusr's** sensitive health and location data scope: $\$50,000 \times 3.75 = 187,500$ SAR (one-time)
- **PDPL Compliance Setup:** The average small business spends \$2,800 per employee per year on cybersecurity [64]. Given that **Yusr** handles sensitive personal health data and GPS location of pilgrims classified as high-risk sensitive data under PDPL — a 45% compliance premium applies [65]: $\$2,800 \times 12 \text{ employees} \times 1.45 \times 3.75 = 182,700$ SAR/year

Total Cybersecurity & Data Protection Compliance Cost (Year 1) = 370,200 SAR

The full breakdown is shown in [Table 12 : Cybersecurity & Data Protection Compliance Costs].

6.1.1.8 Wristband Sanitization Equipment

Since **Yusr's** wristbands are worn directly on pilgrims' skin and shared between multiple users across rental cycles, each distribution booth must be equipped with UV-C sanitization devices to ensure full pathogen elimination between rentals. Given the high-density crowd environment of Al-Masjid Al-Haram and the strict hygiene protocols enforced by Saudi Arabia's Ministry of Health [66], UV-C sanitization is a mandatory operational requirement.

Yusr adopts the PhoneSoap XL UV Sanitizer [67] — a professional-grade UV-C device that sanitizes electronic devices using 360° UV-C light exposure. To ensure continuous and uninterrupted sanitization operations across all rental transactions, 5 units are deployed per booth.

Equipment Cost: $5 \text{ units} \times 5 \text{ booths} \times 1,267.99 \text{ SAR} = 31,699.75 \text{ SAR (one-time)}$

Annual Maintenance (bulb replacement, 10% of purchase price): $31,699.75 \times 10\% = 3,170 \text{ SAR/year}$

Total Wristband Sanitization Equipment Cost (Year 1) = 34,870 SAR

The full breakdown is shown in [Table 13 : Wristband Sanitization Equipment Costs].

6.1.1.9 Government & Legal

Government and legal costs cover mandatory registrations, permits, insurance, and legal services required for **Yusr's** operations.

- **Commercial Registration:** 200 SAR/year as set by the Saudi Ministry of Commerce [68].
- **Distribution Booth Seasonal Permit:** Operating distribution points inside Al-Masjid Al-Haram requires an official seasonal work permit starting at 10,000 SAR [69].
- **Insurance Subscription (GOSI):** **Yusr** is required to enroll its six employees in the General Organization for Social Insurance (GOSI), assuming all employees are Saudi nationals. Based on the official GOSI employer contribution rates [70], the employer contribution is 9% of the monthly salary (Annuities Branch). Applying this to total monthly salaries of 100,016 SAR: $100,016 \times 9\% \times 12 = \approx 108,017.28$ SAR/year
- **Legal Services:** **Yusr** will engage Tahakum Law Firm [71] for contract drafting, review, and regulatory compliance. Package 1 costs 18,000 SAR/year and includes 10 monthly consulting hours and drafting/reviewing 5 contracts per month.

The full breakdown is shown in [Table 14 : Government & Legal Costs] .

6.1.1.10 Distribution Booth Infrastructure

- **Distribution Booths at Al-Masjid Al-Haram:** **Yusr** operates five professional retail distribution booths at the five main gates of Al-Masjid Al-Haram — Bab Al-Malik Abdulaziz, Bab Al-Safa, Bab Al-Fath, Bab Al-Umrah, and Bab Al-Malik Fahd — to ensure year-round product accessibility for pilgrims and visitors throughout all seasons. Each booth consists of a branded professional display counter and a POS payment terminal.
- **Professional Display Counter Rental:** Based on industry benchmarks, annual rental costs approximately one-third of the hardware purchase price [72]. Given a professional branded counter purchase price of \$15,000 ($\approx 56,250$ SAR), the annual rental cost per

counter is approximately 18,750 SAR. For five counters: $18,750 \text{ SAR} \times 5 = 93,750 \text{ SAR/year}$

- **POS Payment Terminal:** Each booth requires a dedicated POS terminal supporting contactless payments. Based on the Saudi Arabia POS Terminal Market Report [73], professional contactless-enabled terminals in Saudi Arabia cost between 800 SAR and 2,500 SAR. Adopting the upper range: $2,500 \text{ SAR} \times 5 = 12,500 \text{ SAR/year}$

Total annual distribution booth infrastructure cost:

$$93,750 \text{ SAR} + 12,500 \text{ SAR} = 106,250 \text{ SAR/year.}$$

The full breakdown is shown in [Table 15 : Distribution Booth Infrastructure Costs].

6.1.1.11 Locations

- **Office Rent:** A 72 sqm furnished office in Riyadh at 116,472 SAR/year [74], covering WiFi, furniture, electricity, and water under a 5-year fixed contract.
- **Storage / Warehouse:** Since Yusr operates a batch production model, storage is required for approximately one month of inventory at a time — approximately 14,140 units ($169,676 \div 12$). This estimate assumes efficient vertical shelving and compact packaging suitable for small electronic devices, requiring approximately 10 m^2 . Based on WheelKeep self-storage rates in Makkah, large climate-controlled storage units cost between 2,500 per month [75]:

$$10 \text{ m}^2 \times 2500 \text{ SAR} \times 12 \text{ months} = 300,000 \text{ SAR/year}$$

The full breakdown is shown in [Table 16 : Locations Costs].

6.1.1.12 Hardware Tools Maintenance

- **Hardware Tools Maintenance:** Annual maintenance costs are estimated for each device procured in Section 2.1.5. Printer maintenance is estimated at 315 SAR/year [76][77]. iPhone maintenance costs 99 SAR/year [78][79]. Laptop maintenance is estimated at 797 SAR/year [80]. The total annual hardware tools maintenance cost is therefore:

$$315 + 99 + 797 = 1,211 \text{ SAR/year}$$

The full breakdown is shown in [Table 17 : Hardware Tools Maintenance CostsTable 17].

[Table 18] summarizes all fixed costs for Year 1.

Table 6 :Employee Salaries

#	Job Title	Monthly Salary (SAR)	Annual Salary (SAR)
1	Software Developer [30]	11,500 SAR	138,000 SAR
2	Website Developer [31]	8,000 SAR	96,000 SAR
3	Hardware Developer [32]	13,133 SAR	157,600 SAR
4	AI Specialist [33]	18,158 SAR	217,900 SAR
5	Customer Support Specialist [34]	5,748 SAR	68,976 SAR
6	Marketing Specialist [35]	7,500 SAR	90,000 SAR
7	Project Manager [36]	19,941 SAR	239,300 SAR
8	Sales Employee [37] ($\times 5$ booths)	4,000 SAR/person	240,000 SAR
Total Monthly Cost =			$\approx 103,980$ SAR
Total Annual Salaries Cost =			$\approx 1,247,776$ SAR

** Note: The annual salary for Hardware Engineer, AI Specialist, and Project Manager is sourced directly from World Salaries [32][33][36] as a published annual figure and may not equal the displayed monthly salary $\times 12$ exactly, due to rounding in the monthly approximation. The annual column values are used as the basis for all cost calculations.*

Table 7: App & Website Development

#	Item	Type	Cost (SAR)
1	Mobile App Development (Complex + AI) [38]	One-Time	475,000 SAR
2	App Store Publishing Fees [39][40]	One-Time	465 SAR
3	App Maintenance & Updates [41]	Annual	83,125 SAR
4	Website Development – e-commerce [42][42]	One-Time	30,000 SAR
5	Website Maintenance [43]	Annual	5,250 SAR
Total Development Cost (Year 1) =			593,840 SAR

Table 8 : Cloud Services & Hosting

#	Service	Provider	Annual Cost (SAR)
1	Amazon EC2 – App Hosting & AI Processing	AWS ME (Bahrain)	\$48.91/month $\approx$ 2,200 SAR/yr
2	Amazon RDS – MySQL Managed Database	AWS	\$165.44/month $\approx$ 7,445 SAR/yr
3	Amazon S3 – Data Storage	AWS	\$0.23/month $\approx$ 10.35 SAR/yr
Total Cloud & Hosting Cost per Year =			$\approx$ 9,655.35 SAR

Table 9 : Marketing & Advertising

#	Marketing Channel	Components	Annual Cost (SAR)
1	Instagram	Ads + 2 Micro + 2 Macro	181,500 SAR
2	X	Ads	13,500 SAR
3	TikTok	Ads + 2 Micro + 2 Macro	227,532 SAR
4	YouTube	Ads + 2 Micro + 2 Macro	352,125 SAR
5	Search Engine Optimization (SEO)	Year-round	120,000 SAR
6	Pilgrimage Platform Ads	Year-round	36,000 SAR
7	Marketing Agency Fee	Year-round	300,000 SAR
Total Annual Marketing Cost =			1,230,657 SAR

Table 10 : Hardware & Software Tools

#	Item	Quantity	Cost (SAR)
1	MacBook Pro 16" M4 Laptops [54]	7	59,493 SAR (One-Time)
2	iPhone for IOS application Testing [55]	1	7,299 SAR (One-Time)
3	Android Phone for android application Testing [56]	1	5,499 SAR (One-Time)
4	iPad Pro 13" M4 [57]	5	27,495 SAR (One-Time)
5	Microsoft 365 Business Basic [58]	7 users	3,938 SAR/year
6	Office Printer – HP LaserJet [59]	1	2,299 SAR (One-Time)
Total Hardware & Software Tools Cost (Year 1) =			106,023 SAR

Table 11 : Beacons Setup Costs

#	Item	Type	Cost (SAR)
1	Hardware Cost – Kontakt.io Anchor Beacon 2 × 2,000 [60]	One-Time	262,500 SAR
2	Installation Labor [61]	One-Time	750,000 SAR
3	Annual Maintenance (10% of hardware cost)	Annual	26,250 SAR
4	Kio Cloud Platform – 2,000 devices	Annual	24,000 SAR
Total Beacons Setup Cost (Year 1) =			1,062,750 SAR

Table 12 : Cybersecurity & Data Protection Compliance Costs

#	Item	Type	Cost (SAR)
1	ISO 27001 Implementation & Certification [63]	One-Time	187,500 SAR
2	PDPL Compliance Setup [64][65]	Annual	182,700 SAR
Total Cybersecurity & Data Protection Cost (Year 1) =			370,200 SAR

Table 13 : Wristband Sanitization Equipment Costs

#	Item	Type	Cost (SAR)
1	PhoneSoap XL UV Sanitizer – 5 units × 5 booths [67]	One-Time	31,699.75 SAR
2	Annual Maintenance (bulb replacement, 10%)	Annual	3,170 SAR
Total Wristband Sanitization Equipment Cost (Year 1) =			34,870 SAR

Table 14 : Government & Legal Costs

#	Item	Cost (SAR)
1	Commercial Registration / Business License [68]	200 SAR
2	Distribution Booth Seasonal Permit – Al-Masjid Al-Haram [69]	10,000 SAR
3	Insurance Subscription – GOSI, 9% of monthly salaries × 12 [70]	108,017 SAR
4	Legal Services – Tahakum Law Firm, Package 1 [71]	18,000 SAR
Total Government & Legal Cost(Year 1) =		136,217 SAR

Table 15 : Distribution Booth Infrastructure Costs

#	Item	Cost (SAR)
1	Professional Display Counter Rental – 5 counters × 18,750 SAR [72]	93,750 SAR
	POS Payment Terminal – 5 terminals × 2,500 SAR [73]	12,500 SAR
Total Distribution Booth Infrastructure Cost (Year 1) =		106,250 SAR

Table 16 :Locations Costs

#	Item	Cost (SAR)
1	Office Rent – 72 sqm, Riyadh [74]	116,472 SAR
2	Storage / Warehouse – Batch production, ≈14,140 units/month, 10 m ² [75]	300,000 SAR
Total Locations Cost (Year 1) =		416,472 SAR

Table 17 : Hardware Tools Maintenance Costs

#	Item	Cost (SAR)
1	Hardware Tools Maintenance – Printer [76][77], Phones [78][79], Laptop [80]	1,211 SAR

Total Hardware Tools Maintenance Cost (Year 1) =	1,211 SAR
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Table 18 : Fixed Costs Summary

Cost Category	Annual Cost (SAR)
Employee Salaries	1,247,776 SAR
App & Website Development	593,840 SAR
Cloud Services / Hosting (AWS)	9,655 SAR
Marketing & Advertising	1,230,657 SAR
Hardware & Software Tools	106,023 SAR
Beacons Setup (Hardware, Installation, Maintenance, Cloud)	1,062,750 SAR
Cybersecurity & Data Protection Compliance	370,200 SAR
Wristband Sanitization Equipment	34,870 SAR
Government & Legal (Registration, Permits, GOSI, Legal Services)	136,217 SAR
Distribution Booth Infrastructure	106,250 SAR
Locations (Office, Storage)	416,472 SAR
Hardware Tools Maintenance	1,211 SAR
Total Fixed Costs (Year 1) =	5,315,921 SAR

6.1.2 Variable Costs

Variable costs are expenditures that fluctuate in proportion to **Yusr's** production volume and transaction activity. **Yusr** has two primary variable costs: hardware manufacturing cost per unit and payment gateway fees.

6.1.2.1 Hardware Components & Manufacturing

The **Yusr** smart wristband integrates nine primary hardware components covering sensing, communication, power, output, and enclosure functions. The total Bill of Materials

(BOM) cost is estimated at 185 SAR per unit based on current market prices. A teardown benchmark of a comparable sensor-based wristband by IHS Teardown Services [81] indicates

that manufacturing and assembly adds \$17.36 USD ($\approx$ 65.10 SAR) per unit, bringing the total hardware cost per unit to approximately 250 SAR. Since this cost varies directly with the number of units produced, it is classified as a variable cost. The component breakdown is shown in [Table 19].

6.1.2.2 Hardware Manufacturing Cost

As detailed in Section 2.2.1, each **Yusr** wristband costs approximately 250 SAR to manufacture. To determine a realistic first-year target, we applied a 5% market penetration rate directly to our SOM, aligned with revenue projection benchmarks for startups which suggest capturing 5% of SOM in the first year as a realistic target [91].

Consumer trends indicate 42.7% prefer purchasing and 57.3% prefer renting [92]. Applying these proportions:

Small wearable device batteries typically sustain 300 charge cycles before noticeable capacity degradation [93].

However, given that operating temperatures in Makkah exceed 45°C — which reduces lithium-ion battery cycle life by up to 50% [94] — combined with the potential for mishandling and rough usage by pilgrims in a high-density crowd environment, the effective lifespan of each sanitization device is conservatively estimated to reduce to approximately 75 operational cycles ($300 \times 50\% \times 50\%$), necessitating frequent replacement and contributing to the annual sanitization cost.

$$\text{Units Sold Y1} = (598,850,187 \times 0.427 \times 0.05) \div 267 = 47,885 \text{ units}$$

$$\text{Rental Fleet} = 428,926 \div 75 = 5,719 \text{ units}$$

$$\text{Total Units Produced Y1} = 47,885 + 5,719 = 53,604 \text{ units}$$

$$\text{Hardware Manufacturing Cost} = 53,604 \times 250 \text{ SAR} \approx 13,401,000 \text{ SAR}$$

6.1.2.3 Wristband Maintenance Cost

Following industry best practices for rental equipment, which recommend scheduling maintenance after every 100 operating hours [95], and given that each rental period spans approximately 24 hours, maintenance is required after every 4 rentals ($4 \times 24 = 96 \approx 100$ hours). Based on 428,926 total rental units, the number of maintenance cycles required annually is:

$$428,926 \div 4 = 107,232 \text{ maintenance cycles}$$

Applying an annual maintenance cost of 20% of the unit manufacturing cost [96], the total wristband maintenance cost is calculated as:

$$107,232 \times (250 \times 20\%) = 107,232 \times 50 \approx 5,361,600 \text{ SAR/year}$$

6.1.2.4 Packaging Cost

Yusr adopts a packaging cost of 10 SAR per package. This estimate is based on a 2026 wholesale custom-box pricing guide[97], which states that custom boxes typically cost \$1.20–\$6.50 per unit, while cardstock/folding-carton packaging typically ranges from \$1.30 to \$3.50 per unit before additional customization.

Since **Yusr** plans to use better branded packaging with printing and finishing rather than basic low-cost packaging, 10 SAR per package is adopted as a reasonable estimate. Based on Year 1 targets of 47,885 sold wristbands and 428,926 rented wristbands, the packaging cost is calculated as follows:

$$\text{Sold units: } 47,885 \times 10 = 478,850 \text{ SAR}$$

$$\text{Rented units: } 428,926 \times 10 = 4,289,260 \text{ SAR}$$

Therefore, the total estimated **Year 1 packaging cost = 4,768,110 SAR.**

6.1.2.5 Payment Gateway Fees

Since **Yusr** serves both local and international pilgrims, a hybrid payment approach is used: local payments via mada and international payments via Visa/Mastercard through Tap

Payments. Based on GASTAT statistics for Umrah (Q1 2025) [98] and Hajj 2024 [99], the customer mix is approximately 48.1% local and 51.9% international. Fee structure:

- **Local payments (mada):** 1% per transaction [100]
- **International payments (Visa/Mastercard):** 3.75% + 1 SAR service fee per transaction [101]

Based on Year 1 targets of 47,885 units sold at 267 SAR and 428,926 units rented at 40 SAR/rental:

$$\text{Local: } (47,885 \times 0.481 \times 267 \times 1\%) + (428,926 \times 0.481 \times 40 \times 1\%) = 61,497.3 + 82,525.4 = 144,022 \text{ SAR}$$

$$\text{Intl \% fee: } (47,885 \times 0.519 \times 267 + 428,926 \times 0.519 \times 40) \times 3.75\% = 582,752 \text{ SAR}$$

$$\text{Intl service fee: } (47,885 + 428,926) \times 0.519 \times 1 \text{ SAR} = 247,464 \text{ SAR}$$

$$\text{Total International} = 582,752 + 247,464 = 830,217 \text{ SAR}$$

$$\text{Total Payment Gateway} = 144,022 + 830,217 = 974,240 \text{ SAR/year}$$

6.1.2.6 Import Customs Duty & Shipping

Since all **Yusr** wristband components are sourced from Chinese and American suppliers and assembled at the manufacturing facility in China prior to shipment, the fully assembled units are imported into Saudi Arabia as finished goods. Given that each wristband contains a Li-Ion Polymer battery [89], which is classified as a Dangerous Good requiring special handling during air transport, **Yusr** adopts Express Courier shipping (via DHL/FedEx) to ensure safe and compliant delivery. Based on industry benchmarks, express courier rates from China to Saudi Arabia range between \$7–\$15 per kg [102]. Adopting \$12/kg as a representative mid-range rate:

- **Shipping Cost:** Wristband weight with packaging $\approx 100\text{g} = 0.1 \text{ kg}$ per unit $53,604 \text{ units} \times 0.1 \text{ kg} \times \$12 \times 3.75 = 241,200 \text{ SAR}$
- **Import Customs Duty:** As a member of the Gulf Cooperation Council (GCC), Saudi Arabia applies a standard customs duty rate of 5% on the CIF value (Cost, Insurance, and Freight) of imported electronic goods [103]. The CIF value per unit is calculated as: $250 \text{ SAR (unit cost)} + 4.5 \text{ SAR (shipping per unit)} = 254.5 \text{ SAR}$

$$53,604 \text{ units} \times 254.5 \text{ SAR} \times 5\% = 682,091 \text{ SAR}$$

Total Import Shipping & Customs Duty (Year 1) $\approx$ 923,291 SAR

All variable costs for Year 1 are summarized in [Table 20]

Table 19 : Hardware Components Specifications and Cost

Component	Product	Source	Unit Price (SAR)
Heart Rate Sensor [82]	MAX30102	SunFounder	$\approx$ 28.84 SAR
Accelerometer / Gyroscope [83]	GY-521 MPU-6050 6DOF	HAOYU	$\approx$ 5.25 SAR
GPS Module [84]	NEO-6M (GY-NEO6MV2)	Adafruit	$\approx$ 40 SAR
Microcontroller [85]	ESP32-WROOM-32	Adafruit	$\approx$ 28.84 SAR
Vibration Motor [86]	Flat Coin 10×3mm DC 3V	SparkFun	$\approx$ 11.06 SAR
7-Segment Display [87]	TM1637 4-Digit 0.56"	eBay	$\approx$ 13.05 SAR
Mini Speaker [88]	8 Ohm 0.5W	Adafruit	$\approx$ 14.82 SAR
Rechargeable Battery [89]	Li-Ion Polymer 3.7V 350mAh	Adafruit	$\approx$ 26.06 SAR
USB Charging Module [90]	TP4056 Micro-USB 5V 1A	HAOYU	$\approx$ 3.00 SAR
Total BOM Cost per Unit =			$\approx$ 185 SAR
Total Hardware Cost per Unit (incl. assembly [32]) =			$\approx$ 250 SAR

Table 20 : Variable Costs – Year 1

#	Cost Item	Calculation Basis	Estimated Cost (SAR)
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1	Hardware Manufacturing Cost	$250 \text{ SAR} \times 53,604 \text{ units produced}$	13,396,712 SAR
2	Wristband Maintenance Cost [95][96]	$(\text{Rent units}/4) \times 250 \times 20\%$	5,361,575 SAR
3	Packaging Cost [97]	$\text{Sold units} + \text{Rent units} \times 8$	4,768,110 SAR
4	Payment Gateway – Local [100]	48.1% of transactions (mada 1%)	144,022.7 SAR
5	Payment Gateway – International [101]	51.9% of transactions (3.75% + 1 SAR/txn)	974,240 SAR
6	Import Customs Duty & Shipping	$(53,604 \times 0.375 \text{ SAR/kg}) + (53,604 \times 254.5 \times 5\%)$	924,096 SAR
Total Variable Costs (Year 1) =			25,424,733 SAR

The growth rates applied in [Table 21] are based on relevant industry benchmarks and assumed project conditions. Some items were assigned a 0% growth rate, indicating fixed or stable costs over the project horizon based on long-term cloud pricing models such as Reserved Instances and Savings Plans, where pricing remains constant during the commitment period.

Table 21: Growth rate over 3 years

Item	Cost per item (SAR)	Growth rate	Year1 (SAR)	Year2 (SAR)	Year3 (SAR)
Employee Salaries					
Salaries	-	3.5%[104]	1,247,776	1,291,448	1,336,649
App & Website Development					
Mobile App Development (Complex + AI)	-	One-Time	475,000	-	-

App Store Publishing Fees	-	One-Time	465	-	-
Website Development	-	One-Time	30,000	-	-
App Maintenance & Updates	-	5.5%[105]	83,125	87,697	92,520
Website Maintenance	-	4%[106]	5,250	5,460	5,678
Cloud Services / Hosting (AWS)					
Cloud Services	-	0% [107]	9,655.35	9,655.35	9,655.35
Marketing & Advertising					
Marketing	-	7.84%[108]	1,230,657	1,327,141	1,431,188
Hardware & Software Tools					
MacBook Pro	-	-	59,493	-	-
iPhone for IOS Testing	-	-	7,299	-	-
Android Phone for android Testing	-	-	5,499	-	-
iPad	-	-	27,495	-	-
Microsoft 365 Business Basic	-	5% [109]	3,938	4,135	4,342
Office Printer	-	-	2,299	-	-
Beacons Setup					
Beacons Hardware Cost	-	-	262,500	-	-
Beacons Installation Labor	-	-	750,000	-	-
Beacons Annual Maintenance	-	2.1% [110]	26,250	26,801	27,364
Beacons Kio Cloud Platform	-	2.1% [110]	24,000	24,504	25,019

Cybersecurity & Data Protection Compliance					
ISO 27001 Implementation & Certification	-	-	187,500	-	-
PDPL Compliance Setup	-	5.79% [111]	182,700	193,278	204,469
Wristband Sanitization Equipment					
Sanitization Equipment PhoneSoap Sanitizer	-	-	31,699.75	-	-
Sanitization Equipment Annual Maintenance	-	2.1% [112]	3,170	3,237	3,305
Government & Legal					
Commercial Registration / Business License	-	0% [113]	200	200	200
Distribution Booth Seasonal Permit	-	0% [113]	10,000	10,000	10,000
GOSI	-	0.5% [114]	108,017	108,557	109,100
Legal Services	-	0% [113]	18,000	18,000	18,000
Distribution Booth Infrastructure					
Display Counter Rental	-	4% [115]	93,750	97,500	101,400
POS Terminal	-	One-Time	12,500	-	-
Locations (Office, Storage)					
Office	-	3.22% [116]	116,472	120,222	124,094
Storage	-	3.22% [116]	300,000	309,660	319,631
Hardware Tools Maintenance					
Hardware Tools Maintenance	-	2.4% [117]	1,211	1,240	1,270
Hardware Components					

Heart Rate Sensor	28.84	10.9% [118]	1,545,939	1,714,447	1,901,321
Accelerometer / Gyroscope	5.25	8% [119]	281,421	303,935	328,249
GPS Module	54	11.5% [120]	2,894,616	3,227,497	3,598,659
Microcontroller	28.84	10% [121]	1,545,939	1,700,533	1,870,587
Vibration Motor	11.06	5% [122]	592,860	622,503	653,628
7-Segment Display	13.05	5% [123]	699,532	734,509	771,234
Mini Speaker	14.82	4.66% [124]	794,411	831,431	870,176
Rechargeable Battery	26.06	21.9% [125]	1,396,920	1,702,846	2,075,769
USB Charging Module	3	15.94% [126]	160,812	186,445	216,165
Manufacturing & Assembly	-	5.2% [127]	3,484,260	3,665,442	3,856,044
Wristband Maintenance Cost					
Wristband Maintenance	-	2.1% [112]	5,361,575	5,474,168	5,589,126
Fulfilment					
Packaging	-	3.16% [128]	4,768,110	4,918,782	5,074,216
Shipping	-	4% [129]	924,096	961,060	999,503
Payment gateway fees					
Local Payment (Mada)	-	1% [130]	144,023	145,463	146,917
International Payment (Visa/MasterCard)	-	1% [130]	830,218	838,520	846,905

6.2 Revenue

Yusr's revenue is generated through two streams: **direct wristband sales and daily rentals**.

6.2.1 First Year Revenue:

To determine a realistic first-year target, we applied a 5% market penetration rate directly to our SOM, aligned with revenue projection benchmarks for startups which suggest capturing 5% of SOM in the first year as a realistic target [131].

The SOM is then split into purchase and rental segments based on consumer adoption behavior. Consumer trends indicate that 57.3% of consumers prefer renting over purchasing well-made products, while the remaining 42.7% prefer purchasing [132]. These proportions are applied to SOM to reflect realistic demand distribution between the two models.

$$\begin{aligned} \text{Units Sold}_{Y1} &= \frac{\text{SOM} \times \text{Purchase Share} \times \text{Year 1 Target \% of SOM}}{\text{Purchase Price}} \\ &= \frac{598,850,187 \times 0.427 \times 0.05}{267} \\ &= 47,885 \text{ wristband} \end{aligned}$$

Given that Umrah rituals are typically completed within 3–6 hours in a single visit [133], and pilgrims primarily require safety and tracking support during active ritual performance, we conservatively assume an average rental duration of 1 day per pilgrim. Accordingly, the base daily rental price of 40 SAR is applied, without applying any multi-day discount tiers.

$$\begin{aligned} \text{Units Rented}_{Y1} &= \frac{\text{SOM} \times \text{Rental Share} \times \text{Year 1 Target \% of SOM}}{\text{Daily Rental Price}} \\ &= \frac{598,850,187 \times 0.573 \times 0.05}{40} \\ &= 428,926 \text{ wristband} \end{aligned}$$

So, revenue in first year represents the total income generated from individual pilgrims through both the direct purchase and daily rental models.

$$\begin{aligned}\text{Revenue}_{Y1} &= (\text{Units Sold}_{Y1} \times \text{Purchase Price}) + (\text{Units Rented}_{Y1} \times \text{Daily Rental Price}) \\ &= (47,885 \times 267) + (428,926 \times 40) \\ &= 12,785,295 + 17,157,040 \\ &= \mathbf{29,942,335}\end{aligned}$$

6.2.2 Second Year Revenue:

While revenue growth can be driven by an increase in the price of units sold or rented, **Yusr** adopts a price-based growth strategy for Years 2 and 3, keeping unit volumes constant while adjusting prices in line with market trends. This approach is supported by two independent market benchmarks:

- **Purchase price:** is increased by 11% annually, aligned with the global smart wearables market CAGR of 11%, as reported by Zion Market Research [134].
- **Rental price:** is increased by 7% annually, consistent with the equipment rental industry revenue growth rate of 7% recorded in 2024, as reported by the American Rental Association (ARA) [135].

Applying these growth rates to Year 1 prices:

- **Purchase price:** $267 \times 1.11 = 296.37$ SAR
- **Rental price:** $40 \times 1.07 = 42.80$ SAR

So, the second-year revenue is calculated as:

$$\begin{aligned}& (47,885 \times 296.37) + (428,926 \times 42.80) \\ &= 14,191,677.45 + 18,358,032.8 \\ &= \mathbf{32,549,710}\end{aligned}$$

6.2.3 Third Year Revenue:

Applying these growth rates to Year 2 prices:

- **Purchase price:** $296.37 \times 1.11 = 328.97$ SAR
- **Rental price:** $42.80 \times 1.07 = 45.80$ SAR

So, the third-year revenue is calculated as:

$$\begin{aligned} & (47,885 \times 328.97) + (428,926 \times 45.80) \\ &= 15,752,761.97 + 19,643,095.10 \\ &= \mathbf{35,395,857} \end{aligned}$$

Yusr's revenue over three years, generated from wristband sales and rentals, is summarized in the following table:

Table 22: Annual revenue growth projection.

Year	Revenue
Year 1	29,942,335 SAR
Year 2	32,549,710 SAR
Year 3	35,395,857 SAR

6.3 Projected Statements

In this section, we will present the financial projected statement, which outlines the projected costs, revenue, and profitability over the first three years of **Yusr's** operations (see

Table 23 , Table 24, and Table 25).

6.3.1 First Year (April 2026–2027)

Table 23: First year projected statement.

Fixed Cost		
#	Cost Item	Annual Cost (SAR)
1	Employee Salaries	1,247,776 SAR
2	App & Website Development	593,840 SAR
3	Cloud Services / Hosting (AWS)	9,655 SAR
4	Marketing & Advertising	1,230,657 SAR
5	Hardware & Software Tools	106,023 SAR
6	Beacons Setup (Hardware, Installation, Maintenance, Cloud)	1,062,750 SAR
7	Cybersecurity & Data Protection Compliance	370,200 SAR
8	Wristband Sanitization Equipment	34,870 SAR
9	Government & Legal (Registration, Permits, GOSI, Legal)	136,217 SAR
10	Distribution Booth Infrastructure	106,250 SAR
11	Locations (Office, Storage)	416,472 SAR
12	Hardware Tools Maintenance	1,211 SAR
	Total Fixed Cost =	5,315,921 SAR
Variable Cost		
#	Cost Item	Estimated Cost
1	Hardware Manufacturing Cost	13,392,712 SAR
2	Wristband Maintenance Cost	5,361,575 SAR
3	Packaging Cost	4,768,110 SAR
4	Payment Gateway – Local	144,023 SAR
5	Payment Gateway – International	830,218 SAR
6	Import Customs Duty & Shipping	924,096 SAR
	Total Variable Cost =	25,424,733 SAR
Final Calculations		
Total Cost (Fixed Cost + Variable Cost) =		30,740,655 SAR
Total Revenue =		29,942,335 SAR
Net Profit (Total Revenue - Total Cost) =		(-798,320 SAR)

6.3.2 Second Year (April 2027–2028)

Table 24: Second year projected statement.

Fixed Cost		
#	Cost Item	Annual Cost (SAR)
1	Employee Salaries	1,291,448 SAR
2	App & Website Development	93,157 SAR
3	Cloud Services / Hosting (AWS)	9,655.35 SAR
4	Marketing & Advertising	1,327,141 SAR
5	Hardware & Software Tools	4,135 SAR
6	Beacons Setup (Hardware, Installation, Maintenance, Cloud)	51,305 SAR
7	Cybersecurity & Data Protection Compliance	193,278 SAR
8	Wristband Sanitization Equipment	3,237 SAR
9	Government & Legal (Registration, Permits, GOSI, Legal)	136,757 SAR
10	Distribution Booth Infrastructure	97,500 SAR
11	Locations (Office, Storage)	429,882 SAR
12	Hardware Tools Maintenance	1,240 SAR
	Total Fixed Cost =	3,638,736 SAR
Variable Cost		
#	Cost Item	Estimated Cost
1	Hardware Manufacturing Cost	14,689,587 SAR
2	Wristband Maintenance Cost	5,474,168 SAR
3	Packaging Cost	4,918,782 SAR
4	Payment Gateway – Local	145,463 SAR
5	Payment Gateway – International	838,520 SAR
6	Import Customs Duty & Shipping	961,060 SAR
	Total Variable Cost =	27,027,581 SAR
Final Calculations		
Total Cost (Fixed Cost + Variable Cost) =		30,666,316 SAR
Total Revenue =		32,549,710 SAR
Net Profit (Total Revenue - Total Cost) =		(+1,883,394 SAR)

6.3.3 Third Year (April 2028–2029)

Table 25: Third year projected statement.

Fixed Cost		
#	Cost Item	Annual Cost (SAR)
1	Employee Salaries	1,336,649 SAR
2	App & Website Development	98,199 SAR
3	Cloud Services / Hosting (AWS)	9,655.35 SAR
4	Marketing & Advertising	1,431,188 SAR
5	Hardware & Software Tools	4,342 SAR
6	Beacons Setup (Hardware, Installation, Maintenance, Cloud)	52,383 SAR
7	Cybersecurity & Data Protection Compliance	204,469 SAR
8	Wristband Sanitization Equipment	3,305 SAR
9	Government & Legal (Registration, Permits, GOSI, Legal)	137,300 SAR
10	Distribution Booth Infrastructure	101,400 SAR
11	Locations (Office, Storage)	443,725 SAR
12	Hardware Tools Maintenance	1,270 SAR
	Total Fixed Cost =	3,823,884 SAR
Variable Cost		
#	Cost Item	Estimated Cost
1	Hardware Manufacturing Cost	16,141,833 SAR
2	Wristband Maintenance Cost	5,589,126 SAR
3	Packaging Cost	5,074,216 SAR
4	Payment Gateway – Local	146,917 SAR
5	Payment Gateway – International	846,905 SAR
6	Import Customs Duty & Shipping	999,503 SAR
	Total Variable Cost =	28,798,500 SAR
Final Calculations		
Total Cost (Fixed Cost + Variable Cost) =		32,622,383 SAR
Total Revenue =		35,395,857 SAR
Net Profit (Total Revenue - Total Cost) =		(+2,773,474 SAR)

6.4 Break-Even Analysis

This section presents the break-even point (BEP) for **Yusr**. The break-even point is the level at which total revenue equals total costs, resulting in neither profit nor loss [136]. It is a key financial metric that indicates when **Yusr** will begin to cover all its costs through sales and rental revenue. The break-even point is calculated using the following formulas:

$$\text{Break even point} = \frac{\text{Total Fixed cost}}{\text{Contribution margin}}$$

Where the Contribution Margin is calculated as [137]:

$$\text{Contribution margin} = \text{Total revenue} - \text{Total Variable cost}$$

- Break-even points for each year (coverage ratio):

1. First year (April 2026-2027):

$$\text{Contribution margin} = 29,942,335 - 25,424,733 = 4,517,602 \text{ SAR}$$

$$\text{Break even point} = \frac{5,315,921}{4,517,602} \approx 1.18$$

This means the **Yusr** must achieve about **118% of its expected revenue** to break even in first Year, indicating that it has not yet reached profitability.

2. Second year (April 2027-2028):

$$\text{Contribution margin} = 32,549,710 - 27,027,581 = 5,522,129 \text{ SAR}$$

$$\text{Break even point} = \frac{3,638,736}{5,522,129} \approx 0.65$$

This means **Yusr** needs to achieve about **65% of its expected revenue** to break even in Year 2, indicating that it has reached profitability.

3. Third year (April 2028-2029):

$$\text{Contribution Margin} = 35,395,857 - 28,798,500 = 6,597,357 \text{ SAR}$$

$$\text{Break even point} = \frac{3,823,884}{6,597,357} \approx 0.58$$

This means **Yusr** needs to achieve about **58% of its expected revenue** to break even in Year 3, indicating that profitability is reached earlier and financial performance continues to improve.

- Break-even points in unit terms for each revenue stream:

Since **Yusr** uses a hybrid revenue model combining direct sales and rentals, the break-even point in unit terms is presented separately for each revenue stream to show the number of units required to contribute toward covering total fixed costs.

Because **Yusr** operates with two revenue streams, total fixed costs are allocated between sales and rentals based on each stream's share of total revenue. This ensures that fixed costs are distributed proportionally according to the contribution of each stream to overall revenue [138].

$$\text{Revenue Share} = \frac{\text{Stream Revenue}}{\text{Total Revenue}}$$

$$\text{Adjusted Fixed Cost} = \text{Total Fixed Costs} \times \text{Revenue Share}$$

$$\text{Contribution Margin per Unit} = \text{Revenue per Unit} - \text{Variable Cost per Unit}$$

$$\text{Break - Even Point (units)} = \frac{\text{Adjusted Fixed Cost}}{\text{Contribution Margin per Unit}}$$

1. First Year (April 2026–2027):

- Sales:

$$\text{Revenue Share} = \frac{12,785,295}{29,942,335} = 42.7\%$$

$$\text{Adjusted Fixed Cost} = 5,315,921 \times 0.427 = 2,269,898 \text{ SAR}$$

$$\text{Contribution Margin} = 267 - 250 = 17 \text{ SAR}$$

$$\text{Break even point} = \frac{2,269,898}{17} \approx 133,523 \text{ units}$$

- Rental:

$$\text{Revenue Share} = \frac{17,157,040}{29,942,335} = 57.3\%$$

$$\text{Adjusted Fixed Cost} = 5,315,921 \times 0.573 = 3,046,023 \text{ SAR}$$

$$\text{Contribution Margin} = 40 - 3.33 = 36.67 \text{ SAR}$$

$$\text{Break even point} = \frac{3,046,023}{36.67} \approx 83,071 \text{ rentals}$$

In the first year, the sales stream did not reach its break-even point, with 47,885 units sold compared to 133,523 required. However, the rental stream exceeded its break-even level, achieving 428,926 rentals versus 83,071 required.

2. Second Year (April 2027–2028):

- Sales:

$$\text{Revenue Share} = \frac{14,191,677}{32,549,710} = 43.6\%$$

$$\text{Adjusted Fixed Cost} = 3,638,736 \times 0.436 = 1,586,488 \text{ SAR}$$

$$\text{Contribution Margin} = 296.37 - 250 = 46.37 \text{ SAR}$$

$$\text{Break even point} = \frac{1,586,488}{46.37} \approx 34,211 \text{ units}$$

- Rental:

$$\text{Revenue Share} = \frac{18,358,033}{32,549,710} = 56.4\%$$

$$\text{Adjusted Fixed Cost} = 3,638,736 \times 0.564 = 2,052,248 \text{ SAR}$$

$$\text{Contribution Margin} = 42.80 - 3.33 = 39.47 \text{ SAR}$$

$$\text{Break even point} = \frac{2,052,248}{39.47} \approx 51,996 \text{ rentals}$$

In the second year, both sales and rental streams exceeded their break-even points, with 47,885 units sold compared to 34,211 required, and 428,926 rentals compared to 51,996 required.

3. Third year (April 2028-2029):

- Sales:

$$\text{Revenue Share} = \frac{15,752,762}{35,395,857} = 44.5\%$$

$$\text{Adjusted Fixed Cost} = 3,823,884 \times 0.445 = 1,701,629 \text{ SAR}$$

$$\text{Contribution Margin} = 328.97 - 250 = 78.97 \text{ SAR}$$

$$\text{Break even point} = \frac{1,701,629}{78.97} \approx 21,547 \text{ units}$$

- Rental:

$$\text{Revenue Share} = \frac{19,643,095}{35,395,857} = 55.5\%$$

$$\text{Adjusted Fixed Cost} = 3,823,884 \times 0.555 = 2,122,255 \text{ SAR}$$

$$\text{Contribution Margin} = 45.80 - 3.33 = 42.47 \text{ SAR}$$

$$\text{Break even point} = \frac{2,122,255}{42.47} \approx 49,969 \text{ rentals}$$

By Year 3, both sales and rental streams comfortably exceeded their break-even points.

6.5 Profitability Analysis

In this section, we calculate the profitability of **Yusr** and determine when it will become profitable. Profitability is a crucial indicator of financial sustainability and growth. A startup is

considered profitable when its total revenue exceeds total costs, resulting in a positive net profit.

To assess when **Yusr** will achieve profitability, we calculate Net Profit using the following formula:

$$\text{Net Profit} = \text{Total Revenue} - \text{Total Costs}$$

$$\text{Where: Total Costs} = \text{Fixed Costs} + \text{Variable Costs}$$

- **First Year (April 2026–2027):**

$$\text{Net Profit} = 29,942,335 - 30,740,655 = -798,320 \text{ SAR}$$

Since **Yusr** recorded a loss in the first year, it has not yet achieved profitability. This outcome is expected, as most hardware-based startups do not generate profits in their initial year of operations. Several factors contribute to this deficit. First, significant one-time expenses are concentrated in Year 1, including mobile application development (475,000 SAR), website development (30,000 SAR), and beacon hardware procurement and installation (1,012,500 SAR), all of which are essential for market entry, branding, and setting up operations. Additionally, limited market penetration and brand recognition at this stage result in lower sales volumes, as consumer awareness and trust in the product are still developing. We see that these factors contribute to the negative net profit observed in the first year.

- **Second Year (April 2027–2028):**

$$\text{Net Profit} = 32,549,710 - 30,666,316 = +1,883,394 \text{ SAR}$$

In the second year, **Yusr** achieves profitability for the first time. The shift to a positive net profit is driven by two key factors. First, the majority of one-time setup costs — including app development, beacon installation, and ISO 27001 certification — do not recur in Year 2, significantly reducing the fixed cost burden from 5,325,921 SAR to 3,638,736 SAR. Second, revenue increases as a result of applying a higher purchase price per wristband and a higher daily rental price compared to Year 1, in line with smart wearables and equipment rental market growth benchmarks. These combined effects — lower costs and growing revenue — result in **Yusr** crossing the profitability threshold in its second year of operations.

- **Third Year (April 2028–2029):**

$$\text{Net Profit} = 35,395,857 - 32,622,383 = +2,773,474 \text{ SAR}$$

By the third year, **Yusr's** profitability strengthens further, with net profit growing from 1,883,394 SAR to 2,773,474 SAR — an increase of 890,080 SAR. This improvement reflects continued revenue growth driven by the compounding effect of annual price increases, while total costs grow at a slower pace due to the absence of major one-time expenditures.

The widening gap between revenue and costs confirms that **Yusr** is on a sustainable and scalable financial trajectory.

6.6 Profitability Growth Rate

Profit Growth Rate is the percentage increase in a company's earnings in a time period compared to the previous time period. It provides valuable insights into the company's financial performance and growth trajectory.

To calculate the profit growth, we subtract the prior year's profit from the current year's profit then divide it by the previous year's profit and multiply by 100 to obtain the percentage and get the Profit Growth Rate:

$$\text{Profit Growth} = \text{Current Profit} - \text{Prior Profit}$$

$$\text{Profit Growth Rate} = (\text{Profit Growth} \div |\text{Prior Profit}|) \times 100$$

**Note: When the prior year records a loss (negative profit) the absolute value of the prior year's profit is used in the denominator. This is because dividing by a negative number would produce a misleading negative growth rate, which contradicts the actual financial improvement. Using the absolute value ensures the result accurately reflects the magnitude of progress achieved.[139]*

- **First (April 2026-2027) and second (April 2027-2028) year:**

1- Calculate the profit growth:

$$\text{Profit Growth} = 1,883,394 - (-798,320) = 2,681,714 \text{ SAR}$$

2- Calculate the profit growth rate:

$$\text{Profit Growth Rate} = 2,681,714 \div |-798,320| \times 100 = \approx 335.9\%$$

The profit growth from the first year to the second year shows a significant improvement. Although **Yusr** recorded a loss in the first year, the transition to a positive profit of 1,883,394 SAR in the second year demonstrates that the business model is financially viable once the initial one-time setup costs are fully absorbed. This is a strong sign that **Yusr** is moving in the right direction.

- **Second (April 2027-2028) and third (April 2028-2029) year:**

1- Calculate the profit growth:

$$\text{Profit Growth} = 2,773,474 - 1,883,394 = \mathbf{890,080 \text{ SAR}}$$

2- Calculate the profit growth rate:

$$\text{Profit Growth Rate} = 890,080 \div 1,883,394 \times 100 = \approx \mathbf{47.2\%}$$

The profit growth from the second year to the third year shows further improvement. The profit increased from 1,883,394 SAR to 2,773,474 SAR, resulting in a growth of 890,080 SAR. The growth rate of 47.2% reflects genuine revenue expansion driven by annual price increases aligned with smart wearables and equipment rental market benchmarks. This positive sign confirms that **Yusr** is continuing to move in the right direction and is building a strong and sustainable financial outlook.

6.7 Summary of the Yusr financial plan

The chart below presents a visual summary of Yusr's projected revenue and cost trends over three years. In the first year, revenue remained below costs due to significant one-time startup expenses, including mobile application development, beacon installation, and ISO 27001 certification. However, as these costs were absorbed, revenue grew steadily while total costs stabilized. By the second year (2027–2028), Yusr reached its break-even point where total revenue surpassed total costs, signaling the beginning of profitability. By the third year, the gap between revenue and costs continued to widen, confirming that Yusr is on a sustainable and scalable financial trajectory.

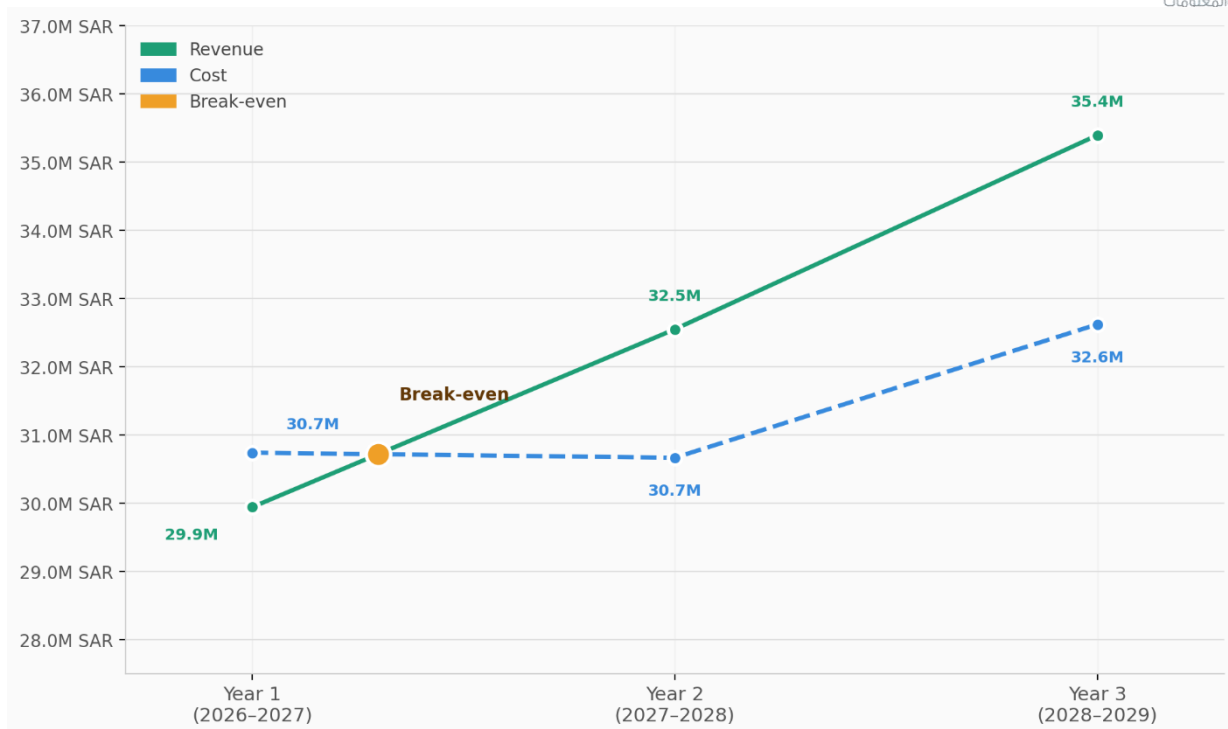


Figure 39: Break-Even Point

7 Our Ask

7.1 Funding Required

To ensure the successful launch and operation of **Yusr**, we have carefully calculated the funding required based on our financial projections. According to our forecast, **Yusr** reaches its break-even point in the second year (2027–2028), with an expected profit of 1,883,394 SAR, meaning the business becomes self-sustaining after the first year of operations.

To determine the funding needed, we applied the Burn Rate metric, which measures the rate at which a startup spends its cash reserves before generating sustainable revenue:

$$\text{Monthly Burn Rate} = \text{Monthly Expenses} - \text{Monthly Revenue} = 30,740,655 \div 12 - 29,942,335 \div 12 = 66,527 \text{ SAR/month}$$

Based on this, the total funding required to sustain **Yusr** through its first year of operations — until it reaches profitability — is:

$$\text{Funding Required} = \text{Monthly Burn Rate} \times 12 \text{ months} = 66,527 \times 12 = 798,320 \text{ SAR}$$

In addition to covering the first-year deficit, industry best practices recommend maintaining an operating reserve of 2–3 months of total monthly expenses as a financial buffer to protect against unexpected costs, supply chain delays, or market fluctuations [140].

Accordingly, **Yusr** incorporates a 3-month operating reserve:

$$\text{Operating Reserve} = (30,740,655 \div 12) \times 3 = 2,561,721 \times 3 = 7,685,164 \text{ SAR}$$

Therefore, the total funding required is:

$$\text{Total Funding} = \text{First-Year Deficit} + \text{Operating Reserve} = 798,320 + 7,685,164 = 8,483,484 \text{ SAR}$$

This funding will be directed toward covering the first-year operational deficit — including hardware manufacturing, beacon infrastructure, marketing, salaries, and platform development — alongside a 3-month operating reserve to ensure financial stability during early operations. This places **Yusr** within the **Seed funding stage**, consistent with industry norms for startups with a validated product entering the market.

7.2 Valuation

Based on **Yusr's** first-year revenue projection and applying a standard startup revenue multiplier, the company's estimated valuation is calculated as follows:

$$\text{Valuation} = \text{Year 1 Revenue} \times \text{Multiplier} = 29,942,335 \times 1 = 29,942,335 \text{ SAR}$$

A multiplier of 1x is applied in accordance with industry benchmarks for hardware and low-margin businesses [141]. As for the equity offered to the investor, seed-stage investors typically receive between 15% and 35% of equity [142]. Based on **Yusr's** funding requirements and valuation, the equity stake is calculated as follows:

$$\text{Equity \%} = \text{Funding} \div \text{Valuation} \times 100 = 8,483,484 \div 29,942,335 \times 100 = 28.3\%$$

Therefore, the investor will receive an equity stake of 28.3%, while the founders retain 71.7% of the company.

8 Conclusion

The development of **Yusr** has been an enriching experience that provided our team with deep understanding of what it takes to turn a creative concept into a viable business venture. Our initial efforts focused on uncovering the core challenges encountered by Hajj and Umrah pilgrims, including insufficient guidance, safety risks, and difficulties staying connected. To test and validate our concept, we developed an MVP consisting of a landing page and an explainer video, which proved invaluable in collecting user feedback and sharpening our value proposition.

Informed by thorough market research, we examined the competitive landscape, formulated a comprehensive marketing strategy, and constructed robust financial models aimed at achieving long-term scalability and sustainability. By applying structured tools such as the Business Model Canvas (BMC) alongside detailed financial forecasts, we established a solid operational and strategic foundation for **Yusr's** continued development.

This journey has imparted critical lessons on the value of building around customer needs, staying responsive to feedback, maintaining sound financial planning, and fostering effective collaboration within the team. We came to appreciate the importance of early-stage validation, meaningful engagement with our target audience, and the agility to refine both the product offering and the overall business approach as circumstances evolve.

As this chapter of **Yusr's** growth comes to a close, we are now focused on attracting investors who align with our mission and can help accelerate key milestones, broadening our market reach, and positioning **Yusr** for a successful commercial launch and enduring growth.

9 References

- [1] "Saudi Arabia acknowledges death of 1,300 pilgrims during Hajj," EFE, Jun. 23, 2024. [Online]. Available: <https://efe.com/latest-news/2024-06-23/saudi-arabia-acknowledges-death-of-1300-pilgrims-during-hajj/>
- [2] Salaam Gateway, "How Saudi Arabia is turning religious tourism into a growth engine," 2025. [Online]. Available: <https://salaamgateway.com/story/how-saudi-arabia-is-turning-religious-tourism-into-a-growth-engine>
- [3] M. Al-Sayed et al., "Smart mobile application for real-time monitoring and prediction of pilgrim crowd activities in Hajj and Umrah using wearable sensors," *Journal of Infection and Public Health*, 2025. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1110016825007963>
- [4] H. Al-Mekhlafi et al., "The Impact of Classical Cardiovascular Risk Factors on Hospitalization and Mortality among Hajj Pilgrims," *PMC*, 2023. [Online]. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC10129415/>
- [5] Arab News, "Inside the Kingdom's digitally powered vision for Hajj," 2025. [Online]. Available: <https://www.arabnews.com/node/2603386/saudi-arabia>
- [6] PA Global, "Hajj 2025 Transformed: The Rise of a Digitally-Driven Pilgrimage," 2025. [Online]. Available: <https://pa-global.com/insights/smart-hajj-2025-digital-transformation/>
- [7] Gulf News, "Over 18 million foreign pilgrims perform Umrah in 2024, up 101% from 2022," 2025. [Online]. Available: <https://gulfnews.com/world/gulf/saudi/over-18-million-foreign-pilgrims-perform-umrah-in-2024-up-101-from-2022-1.500190141>
- [8] Setup in Saudi, "Saudi Vision 2030 Pilgrim Experience Program Explained for Global Businesses," 2024. [Online]. Available: <https://www.setupinsaudi.com/en/saudi-vision-2030-pilgrim-experience-program-explained-for-global-businesses>
- [9] General Authority for Statistics (GASTAT). (2024). *Hajj and Umrah Statistics 2024*. Saudi Arabia. <https://www.stats.gov.sa>

- [10] “How to Price a New Consumer Electronic Product for Profit on Hardware Manufacturing Cost | Cad Crowd.” Accessed: Feb. 15, 2025. [Online]. Available: <https://www.cadcrowd.com/blog/how-to-price-a-new-consumer-electronic-product-for-profit-on-hardware-manufacturing-cost/>
- [11] Amazon. “Activity with 1.1” Slim Thin Fitness Waterproof” (ASIN: B0F6JV3H9Z), product listing (Amazon.sa). Accessed: Feb. 17, 2026. Available at: <https://www.amazon.sa/-/en/Activity-with1-1-Slim-thin-Fitness-Waterproof/dp/B0F6JV3H9Z>
- [12] Amazon. “Fitness Activity Pressure Waterproof Bracelet” (ASIN: B0GDTQXWB4), product listing (Amazon.sa). Accessed: Feb. 16, 2026. Available at: <https://www.amazon.sa/-/en/Fitness-Activity-Pressure-Waterproof-Bracelet/dp/B0GDTQXWB4>
- [13] Amazon. (n.d.). “Fitness Activity Smart Bracelet” (ASIN: B09Z6CRHJ6), product listing (Amazon.sa). Accessed: Feb. 16, 2026. Available at: <https://www.amazon.sa/dp/B09Z6CRHJ6?th=1>
- [14] Rentrax. (2024). “How do you price an equipment item for rental? A guide to fair rates. <https://rentrax.com/blog/how-do-you-price-an-equipment-item-for-rental-a-guide-to-fair-rates/>
- [15] U.S. Chamber of Commerce. (2019). “The Rental Economy Takes Flight.” <https://www.uschamber.com/co/good-company/launch-pad/rental-economy-trend-grows>
- [16] Unified Mobility Platform – Al Haram Transport Services. Available at: <https://carts.alharamain.gov.sa/>
- [17] The Digital Insurer. “BMC: Willingness to adopt wearable devices with behavioural and economic incentives...” <https://www.the-digital-insurer.com/library/willingness-to-adopt-wearable-devices-with-behavioural-and-economic-incentives-by-health-insurance-wellness-programs-results-of-a-us-cross-sectional-survey-with-multiple-consumer-health-vignettes/>
- [18] The Pilgrim App, Apple App Store and Google Play Store. [Online]. Available: <https://apps.apple.com/sa/app/the-pilgrim-app/id6499281188?l=ar>
- [19] Tawaf Counter, Apple App Store. [Online]. Available: <https://apps.apple.com/sa/app/tawaf-counter/id6447226579?l=ar>

[20] Umrah GPS Guide, Apple App Store and Google Play Store. [Online].

Available: <https://apps.apple.com/sa/app/%D8%AF%D9%84%D9%8A%D9%84-%D8%A7%D9%84%D8%B9%D9%85%D8%B1%D8%A9-gps/id6754767732?l=ar>

[21] Al-Mutawwif, Apple App Store and Google Play Store. [Online].

Available: <https://apps.apple.com/sa/app/%D8%A7%D9%84%D9%85%D8%B7%D9%88%D9%81-%D9%85%D9%86%D8%A7%D8%B3%D9%83-%D8%A7%D9%84%D8%AD%D8%AC-%D9%88%D8%A7%D9%84%D8%B9%D9%85%D8%B1%D8%A9/id547963253?l=ar>

[22] Nusuk, Ministry of Hajj and Umrah, Apple App Store and Google Play Store. [Online].

Available: <https://apps.apple.com/sa/app/nusuk-%D9%86%D8%B3%D9%83/id6469515422?l=ar>

[23] EasyWP, "Website Traffic Benchmarks," [online]. Available:

<https://www.easywp.com/blog/website-traffic-benchmarks/>. [Accessed: Apr. 9, 2026].

[24] ScaleMath, "What Is A Good Waitlist Conversion Rate?," [online]. Available:

<https://scalemath.com/blog/what-is-a-good-waitlist-conversion-rate/>. [Accessed: Apr. 9, 2026].

[25] FreshSparks, "How to Set Realistic Social Media Goals for Business Growth," [online].

Available: <https://freshsparks.com/social-media-goals/>. [Accessed: Apr. 9, 2026].

[26] Base44, "Base44 Platform," [online]. Available: <https://app.base44.com/>. [Accessed: Apr. 9, 2026].

[27] Google, "Campaign URL Builder," Google Analytics, [online]. Available: <https://ga-dev-tools.google/campaign-url-builder/>. [Accessed: Apr. 8, 2026].

[28] A. Broetje, "How to determine what Google Analytics Direct / None traffic is from," Odd Dog Media, Aug. 30, 2022. [Online]. Available: <https://odd.dog/blog/how-to-determine-what-google-analytics-directnone-traffic-is-from/>. [Accessed: Apr. 9, 2026].

[29] Ubersuggest, "Ubersuggest - SEO & Keyword Tool," [online]. Available: <https://ubersuggest.com/>. [Accessed: Apr. 9, 2026].

- [30] GulfTalent. "Software Developer Salaries in Saudi Arabia." Accessed: Feb. 13, 2025. [Online]. Available: <https://www.gulftalent.com/saudi-arabia/salaries/softwaredeveloper?mode=related>
- [31] Glassdoor. "Salary: Web Developer in Saudi Arabia 2025." Accessed: Feb. 15, 2025. [Online]. Available: https://www.glassdoor.com/Salaries/saudi-arabia-web-developer-salarySRCH_IL.0%2C12_IN207_KO13%2C26.htm
- [32] WorldSalaries. "Average Computer Hardware Engineer Salary in Saudi Arabia for 2025." Accessed: Feb. 13, 2025. [Online]. Available: <https://worldsalaries.com/average-computer-hardware-engineer-salary-in-saudi-arabia/>
- [33] WorldSalaries. "Average Artificial Intelligence and Machine Learning Specialist Salary in Saudi Arabia for 2025." Accessed: Feb. 13, 2025. [Online]. Available: <https://worldsalaries.com/average-artificial-intelligence-and-machine-learning-specialist-salary-in-saudi-arabia/>
- [34] Paylab Saudi Arabia. Salary Customer Support Specialist. Available: <https://saudiarabia.paylab.com/en/salaryinfo/customer-support/customer-support-specialist>
- [35] GulfTalent. "Marketing Specialist Salaries in Saudi Arabia." Accessed: Feb. 13, 2025. [Online]. Available: <https://www.gulftalent.com/saudi-arabia/salaries/marketing-specialist>
- [36] WorldSalaries. "Average Project Manager Salary in Saudi Arabia for 2025." Accessed: Feb. 13, 2025. [Online]. Available: <https://worldsalaries.com/average-project-manager-salary-in-saudi-arabia/>
- [37] GulfTalent. "Sales Representative Salaries in Saudi Arabia." Accessed: Feb. 13, 2025. [Online]. Available: <https://www.gulftalent.com/saudi-arabia/salaries/sales-representative>
- [38] TrangoTech. How Much Does It Cost to Make an App in Saudi Arabia. 2024. Available: <https://trangotech.com/blog/cost-to-make-an-app-in-saudi-arabia/>
- [39] Apple Inc. App Store Connect – Agreements, Tax, and Banking. Available: <https://developer.apple.com/programs/>
- [40] Google LLC. Google Play Console Registration Fee. Available: <https://support.google.com/googleplay/android-developer/answer/6112435>

- [41] Business of Apps. App Development Cost. 2025. Available: <https://www.businessofapps.com/app-developers/research/app-development-cost/>
- [42] X-360.ai. The cost of website development in Saudi Arabia: What to expect. Available: <https://www.x-360.ai/post/the-cost-of-website-development-in-saudi-arabia-what-to-expect>
- [43] Sieg Partners. How much does it cost to develop an e-commerce website in Saudi Arabia (KSA) 2025? Available: <https://www.siegparkers.com/how-much-does-it-cost-to-develop-an-ecommerce-website-in-saudi-arabia-ksa-2025/>
- [44] Amazon Web Services. AWS Pricing Calculator. Available: <https://calculator.aws>
- [45] LocalCitySolutions. Facebook Marketing Cost in Saudi Arabia 2025. Accessed: Apr. 2025. [Online]. Available: <https://localcitysolutions.com/facebook-marketing-cost-in-saudi-arabia-2025/>
- [46] ClickAnalytic. Influencer Price List 2026. Available: <https://www.clickanalytic.com/influencer-price-list/>
- [47] DemandSage. How Much Do Influencers Make 2026. Available: <https://www.demandsage.com/how-much-do-influencers-make/>
- [48] Circleboom. How Much Do Twitter Ads Cost? 2025. [Online]. Available: <https://circleboom.com/blog/how-much-do-twitter-ads-cost/>
- [49] Business of Apps. TikTok Influencer Marketing Costs. 2024. [Online]. Available: <https://www.businessofapps.com/marketplace/influencer-marketing/research/influencer-marketing-costs/>
- [50] AWISEE. YouTube CPM Rates by Country 2025. [Online]. Available: <https://awisee.com/blog/youtube-cpm-rates/>
- [51] Aly, N. "SEO and SEM Cost in Saudi Arabia: Pricing Guide for 2025." Accessed: Mar. 15, 2025. [Online]. Available: <https://alynoah.com/seo-sem-cost-saudi-arabia/>
- [52] Estimated based on regional digital travel platform advertising benchmarks (Almosafer, Seera).
- [53] Sortlist. (2026). The 10 Best Digital Marketing Agencies in Saudi Arabia. [Online]. Available: <https://www.sortlist.com/digital-marketing/saudi-arabia-sa>

- [54] Jarir Bookstore. MacBook Pro 16" M4. Available: <https://www.jarir.com/apple-macbook-pro-laptops-jpm1422.html?childSku=646488>
- [55] Jarir Bookstore. Apple iPhone 16 Pro Max. Available: <https://www.jarir.com/sa-en/apple-iphone-16-pro-max-smartphones-642678.html>
- [56] Jarir Bookstore. Samsung Galaxy S26 Ultra. Available: <https://www.jarir.com/sa-en/samsung-galaxy-s26-ultra-smartphones-jpm1821.html?childSku=677548>
- [57] Jarir Bookstore. Apple iPad Pro 13 M4. Available: <https://www.jarir.com/sa-en/apple-ipad-pro-13-m4-tablet-pc-jpm1444.html?childSku=635598>
- [58] Microsoft. Microsoft 365 Business Basic – Saudi Arabia. Available: <https://www.microsoft.com/ar/microsoft-365/business/compare-all-microsoft-365-business-products?market=sa>
- [59] Jarir Bookstore. HP LaserJet Printer. Available: <https://www.jarir.com/sa-en/printers.html>
- [60] eBay. TM1637 4-Digit 7-Segment Display. Item No. 196127246241. Available: <https://www.ebay.com/itm/196127246241>
- [61] Kontakt.io. "Smart Beacon Two." Available: https://kontakt.io/blog/smart_beacon_two/
- [62] CMS Law. *Data Protection and Cybersecurity Laws in Saudi Arabia*. Available: <https://cms.law/en/int/expert-guides/cms-expert-guide-to-data-protection-and-cyber-security-laws/saudi-arabia>
- [63] Secureframe. *How Much Does ISO 27001 Certification Cost?* Available: <https://secureframe.com/hub/iso-27001/certification-cost>
- [64] ExecWeb. *Cost of Cybersecurity for Small Businesses in 2025*. Available: <https://www.execweb.com/post/cost-of-cybersecurity-for-small-businesses>
- [65] Total Assure. *Cost of Cybersecurity for Small Businesses in 2025*. Available: <https://www.totalassure.com/blog/Cost-of-Cybersecurity-for-Small-Businesses-in-2025>
- [66] Ministry of Health, Kingdom of Saudi Arabia. *Health Requirements and Guidelines for Pilgrims and Umrah Performers 1446 AH/2025*. Accessed: Apr. 2025. [Online]. Available: <https://www.moh.gov.sa/en/healthawareness/pilgrims-health/pages/default.aspx>

- [67] Amazon Saudi Arabia. *PhoneSoap XL (HomeSoap) UV Light Sanitizer*. Available: <https://amzn.eu/d/01r8NS5R>
- [68] السجل إصدار السعودية. التجارة وزارة التجاري. Available: <https://mc.gov.sa/ar/eservices/Pages/ServiceDetails.aspx?sID=2>
- [69] Al-Saudia Al-Yaum. Business permit fees Makkah Hajj season. 2024. Available: <https://alsaudialyaum.com/news/65260>
- [70] General Organization for Social Insurance (GOSI). Employer Contribution Rates – FAQ. Available: https://www.gosi.gov.sa/GOSIOnline/FAQ_Employer
- [71] Tahakum Law Firm. Legal Services – Riyadh. Available: <https://t7com.sa/>
- [72] Art & Display. (2026). *Trade Show Display Cost: 2026 Guide to Brilliance*. Available: <https://www.artanddisplay.com/trade-show-display-cost/>
- [73] Mordor Intelligence. (2025). *Saudi Arabia POS Terminal Market Report*. Available: <https://www.mordorintelligence.com/industry-reports/saudi-arabia-pos-terminal-market>
- [74] SuperOffice Saudi Arabia. Private Office – Riyadh. Available: <https://superoffice.sa/ar/office-details/private-office-D15-as-sulimaniyah>
- [75] WheeKeep. (2024). *How Much Do Moving Companies Charge for Storage?* Retrieved from <https://wheekeep.com/how-much-do-moving-companies-charge-for-storage/>
- [76] Checktrade. How Much Does Printer Repair Cost in 2025? Accessed: Mar. 15, 2025. [Online]. Available: <https://www.checktrade.com/blog/cost-guides/printer-repair-cost/>
- [77] Advancestuff. Expert Insights: How to Properly Maintain Your Copiers and Printers for Optimal Performance. Accessed: Mar. 15, 2025. [Online]. Available: <https://advancestuff.com/blog/maintain-copiers-printers/>
- [78] Apple Inc. iPhone Repair and Service – Apple Support (SA). Accessed: Mar. 15, 2025. [Online]. Available: <https://support.apple.com/en-sa/iphone/repair>
- [79] Business Research Insights. Smartphone Repair Market Size | Global Analysis [2032]. Accessed: Mar. 15, 2025. [Online]. Available:

<https://www.businessresearchinsights.com/market-reports/smartphone-repair-market-117485>

- [80] Yelp. Cost of Electronics Repair – Last Updated March 2025. Accessed: Mar. 15, 2025. [Online]. Available: https://www.yelp.com/costs/laptop_repair
- [81] IHS Teardown Services. Teardown: Fitbit Flex. Electronics360, 2013. Available: <https://electronics360.globalspec.com/article/3128/teardown-fitbit-flex>
- [82] SunFounder. MAX30102 Heart Rate Sensor Module. Available: <https://www.sunfounder.com/products/max30102-heart-rate-sensor-module-pulse-detection-blood-oxygen-concentration-module>
- [83] HAOYU Electronics. GY-521 MPU-6050 3-Axis Gyroscope Module. Available: <https://www.hotmcu.com/gy521-mpu6050-3axis-acceleration-gyroscope-6dof-module-p-83.html>
- [84] AliExpress. GY-NEO6MV2 NEO-6M GPS Module. Item ID: 1005001635722164. Available: <https://www.adafruit.com/>
- [85] Adafruit Industries. ESP32-WROOM-32. Product ID: 3269. Available: <https://www.adafruit.com/product/3269>
- [86] SparkFun Electronics. Vibration Motor ROB-08449. Available: <https://www.sparkfun.com/products/8449>
- [87] eBay. TM1637 4-Digit 7-Segment Display. Item No. 364270160030. Available: <https://www.ebay.com/itm/364270160030>
- [88] Adafruit Industries. Mono Speaker 1W 8 Ohm. Product ID: 5986. Available: <https://www.adafruit.com/product/5986>
- [89] Adafruit Industries. Li-Ion Polymer Battery 3.7V 350mAh. Product ID: 2750. Available: <https://www.adafruit.com/product/2750>
- [90] HAOYU Electronics. TP4056 Micro-USB Charger Module. Available: <https://www.hotmcu.com/tp4056-micro-usb-5v-1a-lithium-battery-charger-module-p-145.html>
- [91] Airfocus. What is market penetration strategy. Available: <https://airfocus.com/glossary/what-is-market-penetration-strategy/>
- [92] U.S. Chamber of Commerce. Rental economy trend grows. Available: <https://www.uschamber.com/co/good-company/launchpad/rental-economy-trend-grows>

- [93] Battery University. (2020). *BU-808: How to Prolong Lithium-based Batteries*. Retrieved from <https://www.batteryuniversity.com/article/bu-808-how-to-prolong-lithium-based-batteries/>
- [94] Large Battery. (2025). *Lithium Batteries Discharging at High and Low Temperatures*. Retrieved from <https://www.large-battery.com/blog/lithium-batteries-discharging-at-high-and-low-temperatures/>
- [95] EZO. "How to Start an Equipment Rental Business." EZRentOut Blog. Available: <https://ezo.io/ezrentout/blog/equipment-rental-business/>
- [96] Kardex Remstar. *How much should I budget for ASRS maintenance*. Available: <https://info.kardex-remstar.com/knowledge/how-much-should-i-budget-for-asrs-maintenance>
- [97] Yours Packaging. "How Much Do Custom Boxes Cost in 2026? A Complete Wholesale Pricing Guide." Available: <https://yourspackaging.com/how-much-do-custom-boxes-cost-in-2026-a-complete-wholesale-pricing-guide/>
- [98] General Authority for Statistics (GASTAT). Umrah Statistics 2024. Available: <https://www.stats.gov.sa/en/w/news/77>
- [99] General Authority for Statistics (GASTAT). Hajj Statistics Publication 2024. Available: <https://www.stats.gov.sa/documents/20117/2435281/Hajj+Statistics+Publication+2024EN.pdf>
- [100] Tap Payments. Maximize eCommerce profits with mada in Saudi Arabia. Available: <https://blog.tap.company/new-mada-fees-saudi/>
- [101] Tap Payments. Terms and Conditions. 2024. Available: <https://www.tap.company/en-sa/terms-and-conditions>
- [102] Honour Ocean Shipping. *Shipping from China to Saudi Arabia – Complete Guide*. Available: <https://honourocean.com/shipping-from-china-to-saudi-arabia-complete-guide-weekly-loading-container/>
- [103] U.S. International Trade Administration. Saudi Arabia – Import Tariffs. Available: <https://www.trade.gov/country-commercial-guides/saudi-arabia-import-tariffs>
- [104] Willis Towers Watson, "2025 Salary Budget Planning Report," 2025. [Online]. Available: <https://www.wtwco.com/en-cm/insights/2025/07/2025-salary-budget-planning-stability-on-the-surface-strategy-in-the-details>

- [105] Simpalm, “Mobile App Maintenance Guide,” 2024. [Online]. Available: <https://simpalm.com/blog/mobile-app-maintenance-guide>
- [106] Report Prime, “Website Maintenance Services Market,” 2024. [Online]. Available: <https://www.reportprime.com/website-maintenance-services-r14020>
- [107] CloudServus, “Upcoming Microsoft Pricing Changes 2024–2025: What You Need to Know,” 2024. [Online]. Available: <https://www.cloudservus.com/blog/upcoming-microsoft-pricing-changes-for-2024-2025-what-you-need-to-know>
- [108] Market Research Future, “Digital Marketing Software Market,” 2025. [Online]. Available: <https://www.marketresearchfuture.com/reports/digital-marketing-software-market-4938>
- [109] BCS365, “Microsoft Subscription Cost Increase,” 2024. [Online]. Available: <https://bcs365.com/insights/microsofts-five-percent-subscription-cost-increase-how-it-impacts-your-business>
- [110] EIN Presswire, “Consumer Electronics Maintenance Industry Report,” 2025. [Online]. Available: <https://www.einpresswire.com/article/849172704>
- [111] Statista, “Cybersecurity Market Outlook,” 2025. [Online]. Available: <https://www.statista.com/outlook/tmo/cybersecurity/worldwide>
- [112] Research and Markets, “Consumer Electronics Repair and Maintenance Market Report,” 2025. [Online]. Available: <https://www.researchandmarkets.com/report/consumer-electronics-repair-and-maintenance>
- [113] Amazon Web Services (AWS), “AWS Savings Plans and Reserved Instances,” 2025. [Online]. Available: <https://aws.amazon.com/savingsplans/>
- [114] Mercans. "Saudi Arabia – Implementation of New Social Security Law." Statutory Alerts, July 2025. [Online]. Available: <https://mercans.com/resources/statutory-alerts/saudi-arabia-new-social-security-law-3-july-2025/>
- [115] Knight Frank, “Saudi Arabia Retail Market Overview,” 2025. [Online]. Available: <https://thearabianmirror.com/riyadh-mall-rents-rise-retail-demand-grows/>
- [116] Savills, “Global Occupier Markets: Prime Office Costs Q4 2024,” 2025. [Online]. Available: https://www.savills.sa/research_articles/244608/372200-0
- [117] EIN Presswire, “Maintenance Services Market Report,” 2025. [Online]. Available: <https://www.einpresswire.com/article/849172704>

- [118] MarketsandMarkets, "Wearable Medical Device Market," [Online]. Available: <https://www.marketsandmarkets.com/Market-Reports/wearable-medical-device-market-81753973.html>.
- [119] Mordor Intelligence, "MEMS Sensor Market," 2024. [Online]. Available: <https://www.mordorintelligence.com/industry-reports/mems-sensor-market>
- [120] Fortune Business Insights, "GPS Tracking Device Market," 2024. [Online]. Available: <https://www.fortunebusinessinsights.com/gps-tracking-device-market-115344>
- [121] Global Market Insights, "Ultra-Low Power Microcontroller Market," 2024. [Online]. Available: <https://www.gminsights.com/industry-analysis/ultra-low-power-microcontroller-market>
- [122] Global Market Insights, "Vibration Monitoring Market," 2024. [Online]. Available: <https://www.gminsights.com/industry-analysis/vibration-monitoring-market>
- [123] Global Market Insights, "Display Market," 2024. [Online]. Available: <https://www.gminsights.com/industry-analysis/display-market>
- [124] Mordor Intelligence, "Microspeaker Market," 2024. [Online]. Available: <https://www.mordorintelligence.com/industry-reports/microspeaker-market>
- [125] Mordor Intelligence, "Lithium-Ion Battery Market," 2024. [Online]. Available: <https://www.mordorintelligence.com/industry-reports/lithium-ion-battery-market>
- [126] Mordor Intelligence, "Battery Management System Market," 2024. [Online]. Available: <https://www.mordorintelligence.com/industry-reports/battery-management-system-market>
- [127] Industry CRC, "Printed Circuit Board Assembly Market," 2025. [Online]. Available: <https://www.industryarc.com/Report/18291/printed-circuit-boards-pcb-assembly-market.html>
- [128] Zion Market Research, "Packaging Market," ZMR-7962, Nov. 2025. [Online]. Available: <https://www.zionmarketresearch.com/report/packaging-market>. [Accessed: Apr. 21, 2026].
- [129] Market Research Future, "Freight and Logistics Market Research Report – Forecast to 2035," Market Research Future, 2026. [Online]. Available: <https://www.marketresearchfuture.com/reports/freight-and-logistics-market-8698>

- [130] M. O'Leary, "Maximize profits with mada: your guide to lower transaction fees for online payments in Saudi Arabia," *Tap Payments Blog*, Sep. 13, 2023. [Online]. Available: <https://blog.tap.company/new-mada-fees-saudi/>. [Accessed: Apr. 21, 2026].
- [131] FasterCapital. (2025). *Mastering Revenue Projections for Your Startup*. [Online]. Available: <https://fastercapital.com/content/Mastering-Revenue-Projections-for-Your-Startup.html>
- [132] U.S. Chamber of Commerce, "Rental economy trend grows," *CO— by U.S. Chamber of Commerce*. [Online]. Available: <https://www.uschamber.com/co/good-company/launch-pad/rental-economy-trend-grows>
- [133] Pilgrim Hub, "How long is Umrah," *Pilgrim Hub Blog*. [Online]. Available: <https://pilgrimhub.co/blogs/news/how-long-is-umrah>
- [134] Zion Market Research. (2025). *Wearable Technology Market Size, Share, Growth Report, 2034*. Retrieved from <https://www.zionmarketresearch.com/report/wearable-technology-market>
- [135] Rental Management / ARA. (2023). *Equipment rental industry remains resilient*. Retrieved from <https://news.ararental.org/equipment-rental-industry-remains-resilient>
- [136] Corporate Finance Institute, "Break-Even Analysis," Corporate Finance Institute, 2020. [Online]. Available: <https://corporatefinanceinstitute.com/resources/accounting/break-even-analysis/>
- [137] Corporate Finance Institute, "Contribution Margin – Overview," Corporate Finance Institute, 2019. [Online]. Available: <https://corporatefinanceinstitute.com/resources/accounting/contribution-margin-overview/>
- [138] Pennsylvania State University, "Multi-product Break-even Analysis," *Penn State Pressbooks*, 2021. [Online]. Available: <https://psu.pb.unizin.org/acctg211/chapter/multi-product-breakeven-analysis/>
- [139] How to calculate the percentage of increase/decrease with negative numbers – Mathematics Stack Exchange, Mathematics Stack Exchange. [Online].

Available: <https://math.stackexchange.com/questions/716767/how-to-calculate-the-percentage-of-increase-decrease-with-negative-numbers>

[140] Brex, "How to Raise Seed Funding for Your Startup," 2025. [Online].

Available: <https://www.brex.com/spend-trends/startup/seed-funding-for-startups>

[141] "How to Value a Startup: A Guide to Startup Valuations - Capboard." [Online].

Available: <https://www.capboard.io/en/captable/startup-valuations>

[142] Waveup, "Guide to stages of startup funding: From pre-seed to IPO," 2024. [Online].

Available: <https://waveup.com/blog/startup-funding-stages/>